Theory of Commutative Algebra 7

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Contents

1 Why do we take $S = \{1, \bar{x}, \bar{x}^2, \bar{x}^3, \dots\}$? A concrete explanation

Let

$$A = k[x, y]/(xy),$$

and let

$$\bar{x} = x + (xy), \qquad \bar{y} = y + (xy)$$

denote the residue classes of x and y in A.

We want to explain concretely why, when we say that we *localize at* $\bar{x}$, we take

$$S = \{1, \bar{x}, \bar{x}^2, \bar{x}^3, \dots\}.$$

1.1 1. General idea: localization makes chosen elements invertible

If A is a ring and $S \subseteq A$ is a multiplicative set, then the localization $S^{-1}A$ is the ring in which every element of S becomes invertible.

Elements of $S^{-1}A$ are fractions

$$\frac{a}{s}, \qquad a \in A,\ s \in S.$$

So if we want one specific element $u \in A$ to become invertible, the standard choice is to take the smallest multiplicative set containing u, namely

$$S = \{1, u, u^2, u^3, \dots\}.$$

Thus:

- to invert 2 in $\mathbb{Z}$, we take
$$S = \{1, 2, 4, 8, \dots\},$$
and obtain
$$S^{-1}\mathbb{Z} = \mathbb{Z}\left[\frac{1}{2}\right];$$
- to invert x in $k[x]$, we take
$$S = \{1, x, x^2, x^3, \dots\},$$
and obtain
$$S^{-1}k[x] = k[x, x^{-1}];$$
- to invert $\bar{x}$ in $A = k[x, y]/(xy)$, we take
$$S = \{1, \bar{x}, \bar{x}^2, \bar{x}^3, \dots\}.$$

1.2 2. Why not just $\{\bar{x}\}$?

Because localization requires S to be a *multiplicative set*. This means:

- $1 \in S$,
- if $s, t \in S$, then $st \in S$,
- usually one also requires $0 \notin S$.

Now:

- $\{\bar{x}\}$ is not enough, because it does not contain 1;
- $\{1, \bar{x}\}$ is still not enough, because

  $$\bar{x} \cdot \bar{x} = \bar{x}^2$$

  should also lie in S;
- similarly, once $\bar{x}^2$ is in S, then $\bar{x}^3$, $\bar{x}^4$, and so on must also be in S.

Therefore the smallest possible multiplicative set containing $\bar{x}$ is exactly

$$S = \{1, \bar{x}, \bar{x}^2, \bar{x}^3, \dots\}.$$

So when people say informally "localize at $\bar{x}$", this is what they mean.

1.3 3. Why do we write $\bar{x}$ and not x?

In the ring

$$A = k[x, y]/(xy),$$

the elements are not the original polynomials x and y, but their residue classes modulo (xy).

Thus the actual elements of A are

$$\bar{x} = x + (xy), \qquad \bar{y} = y + (xy).$$

So the multiplicative set must live inside A, and therefore the correct notation is

$$S = \{1, \bar{x}, \bar{x}^2, \bar{x}^3, \dots\}.$$

In informal writing one often drops the bar and simply writes

$$S = \{1, x, x^2, \dots\},$$

but the precise version is with $\bar{x}$.

1.4 4. What are the elements of $A = k[x, y]/(xy)$ concretely?

Since $xy = 0$ in the quotient, every monomial containing both x and y vanishes. So every residue class can be represented by something of the form

$$a_0 + f(x) + g(y),$$

more precisely by an expression

$$c + a_1x + a_2x^2 + \cdots + b_1y + b_2y^2 + \cdots$$

with no mixed terms x^iy^j for $i, j \geq 1$.

Thus in A we have

$$\bar{x}\,\bar{y} = 0.$$

This is the key relation.

1.5 5. What happens after localizing at $S = \{1, \bar{x}, \bar{x}^2, \dots\}$?

In $S^{-1}A$, the element $\bar{x}$ becomes invertible. But in A we already have

$$\bar{x}\bar{y} = 0.$$

Since $\bar{x}$ is invertible in the localization, we may multiply both sides by $\bar{x}^{-1}$, obtaining

$$\bar{y} = 0$$

in $S^{-1}A$.

So localizing at $\bar{x}$ forces the whole y-part to disappear.

Therefore the localized ring becomes

$$S^{-1}A \cong k[\bar{x}, \bar{x}^{-1}],$$

the Laurent polynomial ring in $\bar{x}$.

1.6 6. Concrete fractions in the localization

An element of $S^{-1}A$ looks like

$$\frac{a}{\bar{x}^n}, \qquad a \in A.$$

For example:

$$\frac{\bar{x}^3 + \bar{x}}{\bar{x}^2} = \bar{x} + \bar{x}^{-1}.$$

Also,

$$\frac{\bar{y}}{\bar{x}}$$

is formally a fraction, but because $\bar{y} = 0$ in the localization, this fraction is actually 0.

So after localization, everything is expressed only in terms of $\bar{x}$ and its inverse.

1.7 7. Geometric intuition

The ring

$$A = k[x, y]/(xy)$$

is the coordinate ring of the algebraic set

$$V(xy) = V(x) \cup V(y),$$

that is, the union of the two coordinate axes in the plane.

So geometrically we are looking at two lines crossing at the origin:

- the x-axis, given by $y = 0$,
- the y-axis, given by $x = 0$.

Now localizing at $\bar{x}$ means: we restrict to the region where $x \neq 0$. But on the y-axis we have $x = 0$, so that whole component disappears. Only the x-axis away from the origin remains.

This matches the algebraic fact that $\bar{y}$ becomes 0.

1.8 8. Why does localization remove primes that meet S?

A standard theorem says:

$$\operatorname{Spec}(S^{-1}A) \cong \{\mathfrak{p} \in \operatorname{Spec}(A) \mid \mathfrak{p} \cap S = \varnothing\}.$$

So a prime ideal survives after localization if and only if it contains none of the elements of S.

Since

$$S = \{1, \bar{x}, \bar{x}^2, \bar{x}^3, \dots\},$$

this means exactly that a prime survives if and only if it does not contain $\bar{x}$.

1.9 9. Concrete prime ideals in this example

In

$$A = k[x, y]/(xy),$$

the main prime ideals to think about are

$$(\bar{x}), \qquad (\bar{y}), \qquad (\bar{x}, \bar{y}).$$

Now check whether they meet S:

The prime $(\bar{x})$. This ideal contains $\bar{x}$, and $\bar{x} \in S$. Hence

$$(\bar{x}) \cap S \neq \varnothing.$$

So $(\bar{x})$ disappears after localization.

The prime $(\bar{x}, \bar{y})$. This also contains $\bar{x}$, so again

$$(\bar{x}, \bar{y}) \cap S \neq \varnothing.$$

Thus it also disappears.

The prime $(\bar{y})$. This ideal does not contain $\bar{x}$, nor any power of $\bar{x}$. Hence

$$(\bar{y}) \cap S = \varnothing.$$

So $(\bar{y})$ survives.

Under the canonical correspondence

$$\mathrm{Spec}(S^{-1}A) \cong \{\mathfrak{p} \in \mathrm{Spec}(A) \mid \mathfrak{p} \cap S = \varnothing\},$$

the surviving prime is $(\bar{y})$, and in the localized ring it becomes the zero ideal, because $\bar{y} = 0$ there.

Indeed $S^{-1}A \cong k[\bar{x}, \bar{x}^{-1}]$ is a domain, so its only prime corresponding to the generic point is

$$(0).$$

1.10 10. Parallel example from $\mathbb{Z}$

Take

$$A = \mathbb{Z}, \qquad S = \{1, 2, 4, 8, \dots\}.$$

Then

$$S^{-1}\mathbb{Z} = \mathbb{Z}\left[\frac{1}{2}\right].$$

A prime ideal $(p) \subseteq \mathbb{Z}$ survives if and only if it does not meet S.

- (2) meets S, since $2 \in (2)$, so (2) disappears;
- (3), (5), (7), etc. do not meet S, so they survive.

This is exactly the same pattern as in the example with $k[x, y]/(xy)$.

1.11 11. Final summary

To localize at $\bar{x}$, we must choose a multiplicative set containing $\bar{x}$. The smallest one is

$$S = \{1, \bar{x}, \bar{x}^2, \bar{x}^3, \dots\}.$$

This is why S has exactly that form.

In the example

$$A = k[x, y]/(xy),$$

we have

$$\bar{x}\bar{y} = 0.$$

After localizing at S, the element $\bar{x}$ becomes invertible, so necessarily

$$\bar{y} = 0.$$

Hence

$$S^{-1}A \cong k[\bar{x}, \bar{x}^{-1}],$$

and on the level of spectra, localization removes all prime ideals containing $\bar{x}$.

So the slogan is:

$$\boxed{\text{Localizing at } \bar{x} \text{ means inverting } \bar{x}, \text{ hence we use } S = \{1, \bar{x}, \bar{x}^2, \dots\}.}$$

2 The set $D(A)$, the meaning of $V(\mathrm{Ann}(a))$, and minimal primes — concrete explanation

Recall the definition:

$$D(A) = \{\mathfrak{p} \in \mathrm{Spec}(A) \ : \ \exists a \in A \text{ such that } \mathfrak{p} \text{ is minimal in } V(\mathrm{Ann}(a))\}.$$

We now explain carefully what each part means, and then work through concrete examples.

2.1 1. The annihilator $\mathrm{Ann}(a)$

If A is a ring and $a \in A$, then the annihilator of a is

$$\mathrm{Ann}(a) = \{r \in A : ra = 0\}.$$

Thus $\mathrm{Ann}(a)$ is the ideal of all elements of the ring that kill a.

Example. In

$$A = k[x, y]/(xy),$$

let $\bar{x}, \bar{y}$ denote the residue classes of x, y. Since

$$\bar{y}\bar{x} = 0,$$

we have

$$\bar{y} \in \mathrm{Ann}(\bar{x}).$$

In fact one can show

$$\mathrm{Ann}(\bar{x}) = (\bar{y}).$$

2.2 2. The set $V(I)$

If $I \subseteq A$ is an ideal, then

$$V(I) = \{\mathfrak{p} \in \mathrm{Spec}(A) : I \subseteq \mathfrak{p}\}.$$

So $V(I)$ is the set of all prime ideals that contain I.

In particular,

$$V(\mathrm{Ann}(a)) = \{\mathfrak{p} \in \mathrm{Spec}(A) : \mathrm{Ann}(a) \subseteq \mathfrak{p}\}.$$

Thus $V(\mathrm{Ann}(a))$ is the set of all prime ideals containing the annihilator of a.

Geometric remark. In algebraic geometry, $V(I)$ is the closed subset of $\mathrm{Spec}(A)$ defined by I. Algebraically, it is simply the set of primes lying above I.

2.3 3. What does "minimal in $V(\mathrm{Ann}(a))$" mean?

It means:

- first, $\mathfrak{p} \in V(\mathrm{Ann}(a))$, so
$$\mathrm{Ann}(a) \subseteq \mathfrak{p};$$
- second, there is no other prime $\mathfrak{q} \in V(\mathrm{Ann}(a))$ such that
$$\mathfrak{q} \subsetneq \mathfrak{p}.$$

So $\mathfrak{p}$ is *minimal with respect to inclusion* among all prime ideals containing $\mathrm{Ann}(a)$.

Equivalently:

$$\mathfrak{p} \text{ is minimal in } V(\mathrm{Ann}(a)) \quad \Longleftrightarrow \quad \mathfrak{p} \text{ is a minimal prime over } \mathrm{Ann}(a).$$

Important. "Minimal" does *not* mean:

- smallest number of generators,
- smallest dimension,
- smallest cardinality.

It means only: minimal in the partially ordered set of prime ideals, ordered by inclusion.

2.4 4. Example: $A = k[x]$

Let

$$A = k[x], \qquad a = x.$$

Since $k[x]$ is an integral domain and $x \neq 0$, no nonzero element kills x. Hence

$$\mathrm{Ann}(x) = (0).$$

Therefore
$$V(\mathrm{Ann}(x)) = V(0) = \{\mathfrak{p} \in \mathrm{Spec}(k[x]) : (0) \subseteq \mathfrak{p}\}.$$

But every prime ideal contains (0), so

$$V(0) = \mathrm{Spec}(k[x]).$$

Now in the domain $k[x]$, the ideal (0) is prime, and it is the smallest prime ideal by inclusion. Thus (0) is minimal in $V(\mathrm{Ann}(x))$.

So in this example,
$$(0) \in D(k[x]).$$

2.5 5. Example: $A = k[x]/(x^2)$

Let
$$A = k[x]/(x^2),$$

and let $\bar{x}$ be the residue class of x.

Since
$$\bar{x}^2 = 0,$$

the element $\bar{x}$ is nonzero but nilpotent.

Now compute $\mathrm{Ann}(\bar{x})$. Let
$$r = \alpha + \beta\bar{x} \in A.$$

Then
$$r\bar{x} = (\alpha + \beta\bar{x})\bar{x} = \alpha\bar{x} + \beta\bar{x}^2 = \alpha\bar{x}.$$

This is 0 exactly when $\alpha = 0$. Hence

$$\mathrm{Ann}(\bar{x}) = (\bar{x}).$$

So
$$V(\mathrm{Ann}(\bar{x})) = V((\bar{x})) = \{\mathfrak{p} \in \mathrm{Spec}(A) : (\bar{x}) \subseteq \mathfrak{p}\}.$$

But in $k[x]/(x^2)$, the only prime ideal is $(\bar{x})$. Therefore

$$V((\bar{x})) = \{(\bar{x})\}.$$

Hence $(\bar{x})$ is minimal in $V(\mathrm{Ann}(\bar{x}))$, and so

$$(\bar{x}) \in D(A).$$

2.6 6. Example: $A = k[x,y]/(xy)$, with $a = \bar{x}$

Now let
$$A = k[x,y]/(xy),$$
with residue classes $\bar{x}, \bar{y}$.

Because
$$\bar{x}\bar{y} = 0,$$
we know $\bar{y} \in \mathrm{Ann}(\bar{x})$. In fact:
$$\mathrm{Ann}(\bar{x}) = (\bar{y}).$$

Indeed, the elements that kill $\bar{x}$ are precisely the multiples of $\bar{y}$.

Therefore
$$V(\mathrm{Ann}(\bar{x})) = V((\bar{y})) = \{\mathfrak{p} \in \mathrm{Spec}(A) : (\bar{y}) \subseteq \mathfrak{p}\}.$$

So $V((\bar{y}))$ consists of all prime ideals containing $(\bar{y})$. The main ones are
$$(\bar{y}), \qquad (\bar{x}, \bar{y}).$$

Now compare them by inclusion:
$$(\bar{y}) \subsetneq (\bar{x}, \bar{y}).$$

Thus the minimal one is
$$(\bar{y}).$$

So $(\bar{y})$ is minimal in $V(\mathrm{Ann}(\bar{x}))$, hence
$$(\bar{y}) \in D(A).$$

2.7 7. Same ring, but $a = \bar{y}$

Still in
$$A = k[x,y]/(xy),$$
take $a = \bar{y}$.

Since
$$\bar{x}\bar{y} = 0,$$
we get
$$\mathrm{Ann}(\bar{y}) = (\bar{x}).$$

Hence
$$V(\mathrm{Ann}(\bar{y})) = V((\bar{x})) = \{\mathfrak{p} \in \mathrm{Spec}(A) : (\bar{x}) \subseteq \mathfrak{p}\}.$$

The main prime ideals containing $(\bar{x})$ are
$$(\bar{x}), \qquad (\bar{x}, \bar{y}).$$

Again,

$$(\bar{x}) \subsetneq (\bar{x}, \bar{y}),$$

so the minimal one is

$$(\bar{x}).$$

Thus $(\bar{x})$ is minimal in $V(\mathrm{Ann}(\bar{y}))$, and therefore

$$(\bar{x}) \in D(A).$$

2.8 8. Why "minimal" means minimal by inclusion

Take again

$$A = k[x, y]/(xy).$$

Consider

$$V((\bar{y})).$$

This contains at least

$$(\bar{y}), \qquad (\bar{x}, \bar{y}).$$

Since

$$(\bar{y}) \subsetneq (\bar{x}, \bar{y}),$$

the ideal $(\bar{y})$ is the minimal element of this set.

So when we say:

$$(\bar{y}) \text{ is minimal in } V((\bar{y})),$$

we mean only that no strictly smaller prime ideal in that same set lies below it.

This is purely an order-theoretic statement with respect to inclusion.

2.9 9. Another useful example: a domain

Suppose A is an integral domain, and let $a \neq 0$.

Then

$$\mathrm{Ann}(a) = (0),$$

because in a domain, if $ra = 0$ and $a \neq 0$, then $r = 0$.

So

$$V(\mathrm{Ann}(a)) = V(0).$$

The minimal primes over (0) are exactly the minimal prime ideals of the ring. In a domain, (0) is prime, so the only minimal prime over (0) is

$$(0).$$

Thus in a domain,

$$(0) \in D(A).$$

2.10 10. Intuition behind the definition of $D(A)$

The set

$$D(A) = \{\mathfrak{p} \in \operatorname{Spec}(A) \ : \ \exists a \in A \text{ such that } \mathfrak{p} \text{ is minimal in } V(\operatorname{Ann}(a))\}$$

collects those prime ideals that arise as minimal primes over annihilators of elements. So to show that a prime $\mathfrak{p}$ lies in $D(A)$, one must find some element $a \in A$ such that:

1. $\operatorname{Ann}(a) \subseteq \mathfrak{p}$,
2. $\mathfrak{p}$ is minimal among all primes containing $\operatorname{Ann}(a)$.

That is, $\mathfrak{p}$ lies directly above the annihilator of some element.

2.11 11. Compact summary

Recall:

$$D(A) = \{\mathfrak{p} \in \operatorname{Spec}(A) \ : \ \exists a \in A \text{ such that } \mathfrak{p} \text{ is minimal in } V(\operatorname{Ann}(a))\}.$$

Here:

$$\operatorname{Ann}(a) = \{r \in A : ra = 0\},$$

and for any ideal $I \subseteq A$,

$$V(I) = \{\mathfrak{p} \in \operatorname{Spec}(A) : I \subseteq \mathfrak{p}\}.$$

So $V(\operatorname{Ann}(a))$ is the set of prime ideals containing the annihilator of a.
To say that $\mathfrak{p}$ is minimal in $V(\operatorname{Ann}(a))$ means:

$$\mathfrak{p} \in V(\operatorname{Ann}(a))$$

and there is no $\mathfrak{q} \in V(\operatorname{Ann}(a))$ such that

$$\mathfrak{q} \subsetneq \mathfrak{p}.$$

Equivalently, $\mathfrak{p}$ is a minimal prime over $\operatorname{Ann}(a)$.

2.12 12. Short list of the concrete examples

Example 1: $A = k[x]$, $a = x$**.**

$$\operatorname{Ann}(x) = (0), \qquad V(\operatorname{Ann}(x)) = V(0) = \operatorname{Spec}(k[x]),$$

and the minimal prime is

$$(0).$$

Example 2: $A = k[x]/(x^2)$, $a = \bar{x}$.

$$\mathrm{Ann}(\bar{x}) = (\bar{x}), \qquad V(\mathrm{Ann}(\bar{x})) = \{(\bar{x})\},$$

so the minimal prime is

$$(\bar{x}).$$

Example 3: $A = k[x,y]/(xy)$, $a = \bar{x}$.

$$\mathrm{Ann}(\bar{x}) = (\bar{y}), \qquad V(\mathrm{Ann}(\bar{x})) = V((\bar{y})),$$

and the minimal prime is

$$(\bar{y}).$$

Example 4: $A = k[x,y]/(xy)$, $a = \bar{y}$.

$$\mathrm{Ann}(\bar{y}) = (\bar{x}), \qquad V(\mathrm{Ann}(\bar{y})) = V((\bar{x})),$$

and the minimal prime is

$$(\bar{x}).$$

3 The ideal $(x(x-1))$ and the quotient $k[x]/(x(x-1))$

Let k be a field. Consider the polynomial ring $k[x]$.

3.1 1. The ideal $(x(x-1))$

The notation

$$(x(x-1))$$

means the principal ideal generated by the polynomial $x(x-1)$. Thus

$$(x(x-1)) = \{\, f(x)\, x(x-1) \ : \ f(x) \in k[x] \,\}.$$

So $(x(x-1))$ is exactly the set of all polynomials divisible by $x(x-1)$.

Examples of elements in $(x(x-1))$.

$$x(x-1), \qquad x^2(x-1), \qquad (x+3)x(x-1), \qquad x(x-1)(x^2+1).$$

Examples of polynomials not in $(x(x-1))$.

$$x, \qquad x-1, \qquad x^2, \qquad x^2-x+1.$$

Indeed, these are not divisible by $x(x-1)$.

3.2 2. Meaning of the quotient $k[x]/(x(x-1))$

The quotient ring

$$k[x]/(x(x-1))$$

is the ring in which we identify two polynomials if their difference is divisible by $x(x-1)$.

If $\bar{x}$ denotes the residue class of x modulo $(x(x-1))$, then in the quotient we have

$$\bar{x}(\bar{x}-1)=0.$$

Equivalently,

$$\bar{x}^2=\bar{x}.$$

So $\bar{x}$ is an idempotent element.

3.3 3. What do elements of $k[x]/(x(x-1))$ look like?

Every class in $k[x]/(x(x-1))$ has a unique representative of degree < 2. Therefore every element can be written uniquely in the form

$$a+b\bar{x}, \qquad a,b\in k.$$

This comes from polynomial division: for any $f(x)\in k[x]$, there exist $q(x), r(x)\in k[x]$ such that

$$f(x)=q(x)\,x(x-1)+r(x), \qquad \deg r<2.$$

So

$$r(x)=a+bx$$

for some $a,b\in k$, hence

$$f(x)+(x(x-1))=a+b\bar{x}.$$

3.4 4. Multiplication in the quotient

Since

$$\bar{x}^2=\bar{x},$$

we get

$$(a+b\bar{x})(c+d\bar{x})=ac+(ad+bc)\bar{x}+bd\bar{x}^2=ac+(ad+bc+bd)\bar{x}.$$

So multiplication is determined by the relation $\bar{x}^2=\bar{x}$.

3.5 5. Interpretation by values at 0 and 1

A very important fact is that

$$k[x]/(x(x-1))\cong k\times k.$$

Define

$$\Phi : k[x] \longrightarrow k \times k, \qquad \Phi(f) = (f(0), f(1)).$$

This is a ring homomorphism. Its kernel consists of those polynomials f such that

$$f(0) = 0 \qquad \text{and} \qquad f(1) = 0.$$

This means that f is divisible by both x and $x - 1$, hence by $x(x - 1)$. Therefore

$$\ker(\Phi) = (x(x - 1)).$$

Since Φ is surjective, the First Isomorphism Theorem gives

$$k[x]/(x(x - 1)) \cong k \times k.$$

Explicitly, the isomorphism is

$$f(x) + (x(x - 1)) \longmapsto (f(0), f(1)).$$

3.6 6. Images of some important elements

Under the isomorphism

$$k[x]/(x(x - 1)) \cong k \times k,$$

we have

$$1 \longmapsto (1, 1), \qquad \bar{x} \longmapsto (0, 1), \qquad 1 - \bar{x} \longmapsto (1, 0).$$

Hence

$$\bar{x}^2 = \bar{x}, \qquad (1 - \bar{x})^2 = 1 - \bar{x}, \qquad \bar{x}(1 - \bar{x}) = 0.$$

So $\bar{x}$ and $1 - \bar{x}$ are complementary idempotents.

3.7 7. Chinese remainder theorem viewpoint

Since x and $x - 1$ are coprime in $k[x]$, we have

$$(x(x - 1)) = (x) \cap (x - 1),$$

and by the Chinese remainder theorem,

$$k[x]/(x(x - 1)) \cong k[x]/(x) \times k[x]/(x - 1).$$

But

$$k[x]/(x) \cong k, \qquad k[x]/(x - 1) \cong k,$$

so again

$$k[x]/(x(x - 1)) \cong k \times k.$$

3.8 8. Is $k[x]/(x(x-1))$ a domain?

No. In the quotient we have

$$\bar{x}(\bar{x}-1) = 0,$$

but neither $\bar{x}$ nor $\bar{x}-1$ is zero.

Indeed, under the identification with $k \times k$,

$$\bar{x} \mapsto (0,1) \neq (0,0), \qquad \bar{x}-1 \mapsto (-1,0) \neq (0,0).$$

Thus the quotient has zero divisors, so it is not an integral domain.

3.9 9. Prime and maximal ideals

Since

$$k[x]/(x(x-1)) \cong k \times k,$$

its prime ideals correspond to the prime ideals of the two factors. Therefore the quotient has exactly two prime ideals:

$$(x)/(x(x-1)) \qquad \text{and} \qquad (x-1)/(x(x-1)).$$

Both are maximal ideals.

3.10 10. Geometric meaning

The polynomial $x(x-1)$ vanishes exactly at the two points

$$0 \qquad \text{and} \qquad 1.$$

So the quotient

$$k[x]/(x(x-1))$$

is the coordinate ring of the algebraic set

$$\{0,1\} \subseteq \mathbb{A}^1_k.$$

Because the two roots are distinct, the ring is reduced and splits as

$$k \times k.$$

3.11 11. Comparison with $k[x]/(x^2)$

It is useful to compare:

$$k[x]/(x^2) \qquad \text{and} \qquad k[x]/(x(x-1)).$$

In $k[x]/(x^2)$, the class $\bar{x}$ satisfies

$$\bar{x}^2 = 0,$$

so $\bar{x}$ is nilpotent.

In $k[x]/(x(x-1))$, the class $\bar{x}$ satisfies

$$\bar{x}^2 = \bar{x},$$

so $\bar{x}$ is idempotent.

Thus:

- $k[x]/(x^2)$ describes one point with nilpotent thickening;
- $k[x]/(x(x-1))$ describes two distinct reduced points.

3.12 12. Final summary

The ideal

$$(x(x-1)) = \{f(x)x(x-1) : f(x) \in k[x]\}$$

is the set of all polynomials divisible by $x(x-1)$.

The quotient ring

$$k[x]/(x(x-1))$$

is the ring in which

$$\bar{x}(\bar{x}-1) = 0, \qquad \text{equivalently} \qquad \bar{x}^2 = \bar{x}.$$

Every element has a unique form

$$a + b\bar{x}, \qquad a, b \in k,$$

and in fact

$$k[x]/(x(x-1)) \cong k \times k$$

via

$$f(x) + (x(x-1)) \longmapsto (f(0), f(1)).$$

So $k[x]/(x(x-1))$ is the coordinate ring of the two points 0 and 1.

3.13 Why is $\bar{x}$ the class of x, not of $x(x-1)$?

Consider the quotient ring

$$k[x]/(x(x-1)).$$

Let

$$I = (x(x-1)).$$

Then by definition the elements of the quotient $k[x]/I$ are cosets

$$f(x) + I.$$

So every element of the quotient is the class of some polynomial $f(x)$, taken modulo the ideal $(x(x-1))$.

The class of x. The notation

$$\bar{x}$$

means exactly

$$\bar{x} = x + (x(x-1)).$$

So $\bar{x}$ is the residue class of the polynomial x modulo the ideal $(x(x-1))$.

The class of $x(x-1)$. On the other hand, the class of the polynomial $x(x-1)$ is

$$x(x-1) + (x(x-1)).$$

But this is just the zero element of the quotient, because in any quotient ring A/I, every element of the ideal I becomes 0. Since

$$x(x-1) \in (x(x-1)),$$

we get

$$x(x-1) + (x(x-1)) = 0.$$

So:

$$\bar{x} = x + (x(x-1)) \qquad \text{but} \qquad x(x-1) + (x(x-1)) = 0.$$

Thus $\bar{x}$ is *not* the class of $x(x-1)$; it is the class of x.

3.14 Why does this imply $\bar{x}(\bar{x}-1) = 0$?

Since $\bar{x}$ is the class of x, we have

$$\bar{x} - 1 = (x-1) + (x(x-1)).$$

Therefore

$$\bar{x}(\bar{x}-1) = x(x-1) + (x(x-1)) = 0.$$

So in the quotient ring,

$$\bar{x}(\bar{x}-1) = 0.$$

Equivalently,

$$\bar{x}^2 = \bar{x}.$$

3.15 Why does $\bar{x} \mapsto (0,1)$?

Under the isomorphism

$$k[x]/(x(x-1)) \longrightarrow k \times k, \qquad f(x) + (x(x-1)) \longmapsto (f(0), f(1)),$$

the class $\bar{x}$ means the class of the polynomial x. So we evaluate the polynomial x at 0 and 1:

$$x(0) = 0, \qquad x(1) = 1.$$

Hence

$$\bar{x} \longmapsto (0, 1).$$

By contrast, the class of $x(x-1)$ is the zero element in the quotient, so it maps to

$$(0, 0).$$

3.16 A useful analogy

This is exactly like the quotient

$$\mathbb{Z}/6\mathbb{Z}.$$

There:

$$\bar{2} = 2 + 6\mathbb{Z}$$

is the class of 2, while

$$6 + 6\mathbb{Z} = 0$$

is the zero class.

Similarly, in

$$k[x]/(x(x-1)),$$

the class of x is

$$x + (x(x-1)) = \bar{x},$$

whereas the class of the generator of the ideal is

$$x(x-1) + (x(x-1)) = 0.$$

3.17 Final summary

In the quotient ring

$$k[x]/(x(x-1)),$$

we have

$$\bar{x} = x + (x(x-1)),$$

so $\bar{x}$ is the residue class of the polynomial x.

The polynomial $x(x-1)$ itself lies in the ideal $(x(x-1))$, so its class is

$$x(x-1) + (x(x-1)) = 0.$$

Therefore $\bar{x}$ is the class of x, not the class of $x(x-1)$, and this is why

$$\bar{x}(\bar{x} - 1) = 0$$

holds in the quotient.

4 The ideal $\big(x(x-1)(x-2)\big)$ and the quotient $k[x]/\big(x(x-1)(x-2)\big)$

Let k be a field, and consider the polynomial

$$f(x) = x(x-1)(x-2).$$

We now study the ideal

$$\big(x(x-1)(x-2)\big) \subseteq k[x]$$

and the quotient ring

$$k[x]/\big(x(x-1)(x-2)\big).$$

4.1 1. The ideal $\big(x(x-1)(x-2)\big)$

The notation

$$\big(x(x-1)(x-2)\big)$$

means the principal ideal generated by the polynomial $x(x-1)(x-2)$. Thus

$$\big(x(x-1)(x-2)\big) = \{\, g(x)\, x(x-1)(x-2) : g(x) \in k[x] \,\}.$$

So this ideal consists exactly of all polynomials divisible by $x(x-1)(x-2)$.

Examples of elements in the ideal.

$$x(x-1)(x-2), \qquad x^2(x-1)(x-2), \qquad (x+5)x(x-1)(x-2).$$

Examples of polynomials not in the ideal.

$$x, \qquad x(x-1), \qquad x^2, \qquad x^2-x, \qquad x^2-3x+2.$$

Indeed, these are not divisible by $x(x-1)(x-2)$.

4.2 2. The quotient ring

The quotient

$$k[x]/\big(x(x-1)(x-2)\big)$$

is the ring in which two polynomials are identified whenever their difference is divisible by $x(x-1)(x-2)$.

If $\bar{x}$ denotes the residue class of x, then in the quotient we have

$$\bar{x}(\bar{x}-1)(\bar{x}-2) = 0.$$

This is the fundamental relation in the quotient ring.

4.3 3. What do elements of the quotient look like?

Since the polynomial $x(x-1)(x-2)$ has degree 3, every class in the quotient has a unique representative of degree < 3. Therefore every element of

$$k[x]/\big(x(x-1)(x-2)\big)$$

can be written uniquely in the form

$$a + b\bar{x} + c\bar{x}^2, \qquad a, b, c \in k.$$

Indeed, by polynomial division, for every $h(x) \in k[x]$ there exist $q(x), r(x) \in k[x]$ such that

$$h(x) = q(x)\, x(x-1)(x-2) + r(x), \qquad \deg r < 3.$$

Hence

$$h(x) + \big(x(x-1)(x-2)\big) = a + b\bar{x} + c\bar{x}^2.$$

So as a k-vector space, the quotient has basis

$$1,\ \bar{x},\ \bar{x}^2.$$

4.4 4. The relation satisfied by $\bar{x}$

Expand:

$$x(x-1)(x-2) = x^3 - 3x^2 + 2x.$$

Hence in the quotient,

$$\bar{x}^3 - 3\bar{x}^2 + 2\bar{x} = 0,$$

that is,

$$\bar{x}^3 = 3\bar{x}^2 - 2\bar{x}.$$

So every higher power of $\bar{x}$ can be reduced to an expression involving only

$$1,\ \bar{x},\ \bar{x}^2.$$

4.5 5. Geometric meaning

The polynomial

$$x(x-1)(x-2)$$

vanishes at the points

$$0, \qquad 1, \qquad 2.$$

So, when these are distinct in the field k, the quotient ring

$$k[x]/\big(x(x-1)(x-2)\big)$$

is the coordinate ring of the three-point algebraic set

$$\{0, 1, 2\} \subseteq \mathbb{A}^1_k.$$

Thus elements of the quotient may be thought of as polynomial functions restricted to the three points $0, 1, 2$.

4.6 6. Evaluation map and the product ring $k \times k \times k$

Assume now that $0, 1, 2$ are distinct in k, equivalently

$$\mathrm{char}(k) \neq 2.$$

Define

$$\Phi : k[x] \longrightarrow k \times k \times k, \qquad \Phi(h) = (h(0), h(1), h(2)).$$

This is a ring homomorphism.

Its kernel consists of those polynomials $h(x)$ such that

$$h(0) = 0, \qquad h(1) = 0, \qquad h(2) = 0.$$

That means $h(x)$ is divisible by x, by $x - 1$, and by $x - 2$. Since these factors are pairwise coprime, this is equivalent to divisibility by their product:

$$h(x) \in \Big(x(x-1)(x-2)\Big).$$

So

$$\ker(\Phi) = \Big(x(x-1)(x-2)\Big).$$

The map is also surjective, because for any triple

$$(a, b, c) \in k \times k \times k$$

there exists a polynomial of degree at most 2 taking these values at $0, 1, 2$ (Lagrange interpolation).

Therefore, by the First Isomorphism Theorem,

$$k[x]/\Big(x(x-1)(x-2)\Big) \cong k \times k \times k.$$

The isomorphism is explicitly

$$h(x) + \Big(x(x-1)(x-2)\Big) \longmapsto (h(0), h(1), h(2)).$$

4.7 7. Images of some important elements

Under the isomorphism

$$k[x]/\Big(x(x-1)(x-2)\Big) \cong k \times k \times k,$$

we get:

The class of 1. Since

$$1(0) = 1, \qquad 1(1) = 1, \qquad 1(2) = 1,$$

we have

$$1 \longmapsto (1, 1, 1).$$

The class of x. Since

$$x(0) = 0, \qquad x(1) = 1, \qquad x(2) = 2,$$

we get

$$\bar{x} \longmapsto (0, 1, 2).$$

The class of $x - 1$. Since

$$(x-1)(0) = -1, \qquad (x-1)(1) = 0, \qquad (x-1)(2) = 1,$$

we get

$$\bar{x} - 1 \longmapsto (-1, 0, 1).$$

The class of $x - 2$. Since

$$(x-2)(0) = -2, \qquad (x-2)(1) = -1, \qquad (x-2)(2) = 0,$$

we get

$$\bar{x} - 2 \longmapsto (-2, -1, 0).$$

The class of $x(x-1)(x-2)$. This polynomial lies in the ideal, so its class is zero:

$$x(x-1)(x-2) + \big(x(x-1)(x-2)\big) = 0,$$

and under the product-ring description it maps to

$$(0, 0, 0).$$

4.8 8. The Chinese remainder theorem viewpoint

Again assume $\operatorname{char}(k) \neq 2$. Then the ideals

$$(x), \qquad (x-1), \qquad (x-2)$$

are pairwise comaximal, and

$$\big(x(x-1)(x-2)\big) = (x) \cap (x-1) \cap (x-2).$$

Therefore, by the Chinese remainder theorem,

$$k[x]/\big(x(x-1)(x-2)\big) \cong k[x]/(x) \times k[x]/(x-1) \times k[x]/(x-2).$$

But

$$k[x]/(x) \cong k, \qquad k[x]/(x-1) \cong k, \qquad k[x]/(x-2) \cong k,$$

so

$$k[x]/\big(x(x-1)(x-2)\big) \cong k \times k \times k.$$

4.9 9. Idempotents corresponding to the three points

In $k \times k \times k$, the standard idempotents are

$$(1,0,0), \qquad (0,1,0), \qquad (0,0,1).$$

These correspond to special elements of the quotient ring. Using Lagrange interpolation, define

$$\begin{aligned} e_0 &= \frac{(x-1)(x-2)}{(0-1)(0-2)} = \frac{(x-1)(x-2)}{2}, \\ e_1 &= \frac{x(x-2)}{(1-0)(1-2)} = -x(x-2), \\ e_2 &= \frac{x(x-1)}{(2-0)(2-1)} = \frac{x(x-1)}{2}. \end{aligned}$$

Then, in the quotient ring,

$$e_0 \longmapsto (1,0,0), \qquad e_1 \longmapsto (0,1,0), \qquad e_2 \longmapsto (0,0,1).$$

Hence

$$\begin{gathered} e_0^2 = e_0, \qquad e_1^2 = e_1, \qquad e_2^2 = e_2, \\ e_ie_j = 0 \quad (i \neq j), \qquad e_0 + e_1 + e_2 = 1. \end{gathered}$$

So the quotient ring splits into three orthogonal pieces, one for each point $0, 1, 2$.

4.10 10. Is the quotient a domain?

No. Since

$$k[x]/\Big(x(x-1)(x-2)\Big) \cong k \times k \times k,$$

it has many zero divisors.

For example, in the quotient ring,

$$\bar{x}(\bar{x}-1)(\bar{x}-2) = 0,$$

but none of the factors is zero.

Indeed, under the isomorphism:

$$\bar{x} \longmapsto (0,1,2) \neq (0,0,0),$$

$$\begin{aligned} \bar{x} - 1 &\longmapsto (-1,0,1) \neq (0,0,0), \\ \bar{x} - 2 &\longmapsto (-2,-1,0) \neq (0,0,0). \end{aligned}$$

So there are nonzero elements whose product is zero.

4.11 11. Prime and maximal ideals

When $\mathrm{char}(k) \neq 2$, the quotient ring is isomorphic to

$$k \times k \times k.$$

Therefore its prime ideals are exactly the kernels of the three coordinate projections:

$$k \times k \times k \to k.$$

Correspondingly, the prime ideals of

$$k[x]/\big(x(x-1)(x-2)\big)$$

are

$$(x)\Big/\big(x(x-1)(x-2)\big),$$
$$(x-1)\Big/\big(x(x-1)(x-2)\big),$$
$$(x-2)\Big/\big(x(x-1)(x-2)\big).$$

Each is maximal.

So geometrically the spectrum consists of three points.

4.12 12. Comparison with earlier quotients

Comparison with $k[x]/(x^2)$. In $k[x]/(x^2)$, the class $\bar{x}$ satisfies

$$\bar{x}^2 = 0,$$

so $\bar{x}$ is nilpotent. This ring describes one nonreduced point.

Comparison with $k[x]/(x(x-1))$. In $k[x]/(x(x-1))$, the class $\bar{x}$ satisfies

$$\bar{x}(\bar{x}-1) = 0,$$

and the ring splits as

$$k \times k,$$

describing two reduced points.

Now for $k[x]/(x(x-1)(x-2))$. We have

$$\bar{x}(\bar{x}-1)(\bar{x}-2) = 0,$$

and the ring splits as

$$k \times k \times k,$$

describing three reduced points, provided the roots $0, 1, 2$ are distinct.

4.13 13. Important characteristic-2 remark

If $\mathrm{char}(k) = 2$, then $2 = 0$, so
$$x - 2 = x.$$

Hence
$$x(x-1)(x-2) = x(x-1)x = x^2(x-1).$$

So in characteristic 2, the roots are not three distinct points. The quotient ring

$$k[x]/(x^2(x-1))$$

has a repeated factor and is no longer isomorphic to $k \times k \times k$. Instead, it has a nonreduced component.

Thus the decomposition

$$k[x]/\big(x(x-1)(x-2)\big) \cong k \times k \times k$$

holds exactly when the three roots are distinct, i.e. when $\mathrm{char}(k) \neq 2$.

4.14 14. Final summary

The ideal
$$\big(x(x-1)(x-2)\big)$$
consists of all polynomials divisible by $x(x-1)(x-2)$.

The quotient ring
$$k[x]/\big(x(x-1)(x-2)\big)$$
is the ring in which
$$\bar{x}(\bar{x}-1)(\bar{x}-2) = 0.$$

Every element has a unique form

$$a + b\bar{x} + c\bar{x}^2.$$

If $0, 1, 2$ are distinct in k (equivalently $\mathrm{char}(k) \neq 2$), then

$$k[x]/\big(x(x-1)(x-2)\big) \cong k \times k \times k$$

via
$$h(x) + \big(x(x-1)(x-2)\big) \longmapsto (h(0), h(1), h(2)).$$

So this quotient ring is the coordinate ring of the three points

$$0,\ 1,\ 2.$$

5 Coordinate rings of the quotients (x), $(x(x-1))$, $(x(x-1)(x-2))$

Let k be a field. We work in the polynomial ring

$$k[x].$$

For an ideal $I \subseteq k[x]$, the quotient ring

$$k[x]/I$$

is interpreted as the *coordinate ring* of the algebraic set cut out by I in the affine line $\mathbb{A}^1_k$. If $I = (f)$, then the corresponding zero set is

$$V(f) = \{a \in k : f(a) = 0\}.$$

5.1 1. The case $k[x]/(x)$

Consider the ideal

$$(x).$$

It is generated by the polynomial x, so its zero set is

$$V(x) = \{0\}.$$

There is a natural isomorphism

$$k[x]/(x) \cong k,$$

given by evaluation at $x = 0$:

$$f(x) + (x) \longmapsto f(0).$$

Interpretation. In the quotient $k[x]/(x)$, two polynomials are equal if and only if they have the same value at the point 0. Thus this quotient remembers only the value of a polynomial at one point.

Geometric conclusion. The ring

$$k[x]/(x)$$

is the coordinate ring of the single point $\{0\} \subseteq \mathbb{A}^1_k$. This is a reduced point, since the ring has no nonzero nilpotent elements.

5.2 2. The case $k[x]/(x(x-1))$

Now consider the ideal

$$(x(x-1)).$$

The polynomial $x(x-1)$ vanishes exactly at the points

$$0 \quad \text{and} \quad 1.$$

Hence

$$V(x(x-1)) = \{0,1\}.$$

If $0 \neq 1$ in the field k, then the factors x and $x-1$ are coprime, so by the Chinese remainder theorem,

$$k[x]/(x(x-1)) \cong k[x]/(x) \times k[x]/(x-1) \cong k \times k.$$

This isomorphism can be written explicitly as

$$f(x) + (x(x-1)) \longmapsto (f(0), f(1)).$$

Interpretation. An element of the quotient is therefore determined exactly by the pair of values of the polynomial at the points 0 and 1. So this ring is the ring of functions on the two-point set $\{0,1\}$.

Geometric conclusion. The ring

$$k[x]/(x(x-1))$$

is the coordinate ring of the two points

$$\{0,1\} \subseteq \mathbb{A}^1_k.$$

These are two reduced and separated points.

5.3 3. The case $k[x]/(x(x-1)(x-2))$

Now consider the ideal

$$(x(x-1)(x-2)).$$

The polynomial

$$x(x-1)(x-2)$$

vanishes at the points

$$0, \quad 1, \quad 2.$$

So

$$V(x(x-1)(x-2)) = \{0,1,2\}.$$

If the numbers $0,1,2$ are distinct in the field k, for example when $\mathrm{char}(k) \neq 2$, then the factors

$$x, \qquad x-1, \qquad x-2$$

are pairwise coprime. Therefore, by the Chinese remainder theorem,

$$k[x]/(x(x-1)(x-2)) \cong k[x]/(x) \times k[x]/(x-1) \times k[x]/(x-2) \cong k \times k \times k.$$

The explicit isomorphism is

$$f(x) + (x(x-1)(x-2)) \longmapsto (f(0), f(1), f(2)).$$

Interpretation. In this quotient, two polynomials are identified if and only if they have the same values at the points $0, 1, 2$. Thus the ring

$$k[x]/(x(x-1)(x-2))$$

is the ring of functions on the three-point set $\{0, 1, 2\}$.

Geometric conclusion. This is the coordinate ring of the three points

$$\{0, 1, 2\} \subseteq \mathbb{A}^1_k,$$

provided these points are distinct in the field k.

5.4 4. The general pattern

If $a_1, \ldots, a_n \in k$ are pairwise distinct, then

$$k[x]/\Big((x-a_1)\cdots(x-a_n)\Big) \cong k \times \cdots \times k \quad (n \text{ times}),$$

so it is the coordinate ring of the finite set of points

$$\{a_1, \ldots, a_n\} \subseteq \mathbb{A}^1_k.$$

Thus:

$$\begin{aligned} k[x]/(x) \quad &\leftrightarrow \quad \{0\}, \\ k[x]/(x(x-1)) \quad &\leftrightarrow \quad \{0, 1\}, \\ k[x]/(x(x-1)(x-2)) \quad &\leftrightarrow \quad \{0, 1, 2\}. \end{aligned}$$

5.5 5. Important remark: the set of points versus the ring structure

One must distinguish the zero set itself from the full coordinate ring.

For example,

$$k[x]/(x^2)$$

has the same zero set as $k[x]/(x)$, namely

$$V(x^2) = V(x) = \{0\},$$

but it is not the same ring. In $k[x]/(x^2)$, the class $\bar{x}$ satisfies

$$\bar{x}^2 = 0,$$

so it is a nonzero nilpotent element. This means that the quotient describes not an ordinary point, but a *thickened point*.

By contrast, in the cases

$$k[x]/(x), \qquad k[x]/(x(x-1)), \qquad k[x]/(x(x-1)(x-2))$$

with distinct roots, we obtain reduced rings, so they correspond to ordinary separate points.

5.6 6. Summary

- $$k[x]/(x) \cong k$$
  is the coordinate ring of the point $\{0\}$.
- $$k[x]/(x(x-1)) \cong k \times k$$
  is the coordinate ring of the two points $\{0, 1\}$.
- $$k[x]/(x(x-1)(x-2)) \cong k \times k \times k$$
  is the coordinate ring of the three points $\{0, 1, 2\}$, provided $0, 1, 2$ are distinct in k.

In other words, these quotient rings describe polynomial functions restricted respectively to one, two, or three points of the affine line.

6 Ideals, Quotients, Annihilators, Associated Primes, Minimal Primes, and $D(f)$ for $k[x]$ and $k[x, y]$

Convention. Throughout, k is a field, all rings are commutative with 1, and all modules are unital. When we write $\mathrm{Ass}(A/I)$, we mean $\mathrm{Ass}_A(A/I)$, the associated primes of the A-module A/I.

6.1 1. Basic definitions

Ideal. An ideal $I \subseteq A$ is a subset such that:

$$0 \in I, \qquad a, b \in I \Rightarrow a - b \in I, \qquad r \in A,\ a \in I \Rightarrow ra \in I.$$

Examples in $k[x, y]$:

$$(x), \quad (y), \quad (xy), \quad (x^2), \quad (x, y), \quad (x^2, xy).$$

Quotient ring. If $I \subseteq A$ is an ideal, then

$$A/I$$

is the ring of residue classes $a + I$, often written $\overline{a}$. Two elements $a, b \in A$ represent the same class iff

$$a - b \in I.$$

So in A/I, every element of I becomes 0.

Annihilator of an element. If M is an A-module and $m \in M$, then

$$\mathrm{Ann}_A(m) = \{a \in A : am = 0\}.$$

This is an ideal of A.

Annihilator of a module.

$$\mathrm{Ann}_A(M) = \{a \in A : aM = 0\}.$$

In particular, for a quotient module A/I,

$$\mathrm{Ann}_A(A/I) = I.$$

Associated prime. A prime ideal $\mathfrak{p} \subseteq A$ is called an associated prime of M if there exists some $m \in M$ such that

$$\mathrm{Ann}_A(m) = \mathfrak{p}.$$

The set of all associated primes is denoted

$$\mathrm{Ass}_A(M).$$

Minimal primes over an ideal. A prime ideal $\mathfrak{p}$ is minimal over I if

$$I \subseteq \mathfrak{p}$$

and there is no smaller prime ideal containing I. The set of minimal primes over I is denoted

$$\mathrm{Min}(I)$$

or, equivalently, $\mathrm{Min}(A/I)$.

Support. For an A-module M,

$$\mathrm{Supp}_A(M) = \{\mathfrak{p} \in \mathrm{Spec}(A) : M_{\mathfrak{p}} \neq 0\}.$$

For a quotient module A/I,

$$\mathrm{Supp}_A(A/I) = V(I) = \{\mathfrak{p} \in \mathrm{Spec}(A) : I \subseteq \mathfrak{p}\}.$$

Distinguished open set $D(f)$. For $f \in A$,

$$D(f) = \{\mathfrak{p} \in \mathrm{Spec}(A) : f \notin \mathfrak{p}\}.$$

It is the complement of $V(f)$:

$$D(f) = \mathrm{Spec}(A) \setminus V(f).$$

Also,

$$D(fg) = D(f) \cap D(g).$$

6.2 2. The ring $k[x]$

Ideals in $k[x]$. Since $k[x]$ is a principal ideal domain, every ideal is of the form

$$(f(x))$$

for some polynomial $f(x) \in k[x]$.

Examples:

$$(0), \quad (x), \quad (x^2), \quad (x(x-1)), \quad (x^3+1).$$

Prime ideals in $k[x]$. They are:

$$(0)$$

and

$$(f)$$

where f is irreducible.

If k is algebraically closed, then irreducibles are linear, so prime ideals are

$$(0), \qquad (x-a) \text{ for } a \in k.$$

Minimal primes of $k[x]$. Because $k[x]$ is a domain, its only minimal prime is

$$(0).$$

Thus

$$\mathrm{Min}(k[x]) = \{(0)\}.$$

Associated primes of $k[x]$ as a module over itself. Since

$$\mathrm{Ann}_{k[x]}(1) = (0)$$

and (0) is prime, we get

$$\mathrm{Ass}_{k[x]}(k[x]) = \{(0)\}.$$

6.3 3. Example: $k[x]/(x^2)$

Let

$$A = k[x], \qquad M = A/(x^2).$$

Elements of the quotient. Modulo (x^2), every polynomial is congruent to one of the form

$$a + bx.$$

So

$$k[x]/(x^2) = \{a + b\overline{x} : a, b \in k,\ \overline{x}^2 = 0\}.$$

Here $\overline{x} \neq 0$, but $\overline{x}^2 = 0$.

Annihilator of the module.

$$\mathrm{Ann}_A(M) = \mathrm{Ann}_A(A/(x^2)) = (x^2).$$

Annihilator of $\overline{x}$. We compute

$$\mathrm{Ann}_A(\overline{x}) = \{f(x) \in k[x] : f(x)\overline{x} = 0\}.$$

This means

$$f(x)x \in (x^2),$$

which happens exactly when $f(x)$ is divisible by x. Hence

$$\operatorname{Ann}_A(\overline{x}) = (x).$$

Therefore

$$(x) \in \operatorname{Ass}_A(A/(x^2)).$$

In fact,

$$\operatorname{Ass}_A(A/(x^2)) = \{(x)\}.$$

Minimal primes. Since

$$\sqrt{(x^2)} = (x),$$

the only minimal prime over (x^2) is

$$(x).$$

Thus

$$\operatorname{Min}(A/(x^2)) = \{(x)\}.$$

So in this example,

$$\operatorname{Ass}_A(A/(x^2)) = \operatorname{Min}(A/(x^2)) = \{(x)\}.$$

6.4 4. Example: $k[x]/(x(x-1))$

Let

$$A = k[x], \qquad M = A/(x(x-1)).$$

Minimal primes. Since

$$(x(x-1)) = (x) \cap (x-1)$$

in $k[x]$, the minimal primes are

$$(x), \qquad (x-1).$$

Therefore

$$\operatorname{Min}(A/(x(x-1))) = \{(x), (x-1)\}.$$

Associated primes. By the Chinese remainder theorem,

$$k[x]/(x(x-1)) \cong k[x]/(x) \times k[x]/(x-1) \cong k \times k.$$

Hence the associated primes are exactly

$$\operatorname{Ass}_A(A/(x(x-1))) = \{(x), (x-1)\}.$$

Concrete annihilators. The class $\overline{x}$ has annihilator

$$\operatorname{Ann}_A(\overline{x}) = (x-1),$$

and the class $\overline{x-1}$ has annihilator

$$\operatorname{Ann}_A(\overline{x-1}) = (x).$$

6.5 5. The ring $k[x, y]$

Unlike $k[x]$, not every ideal is principal.

Examples of ideals.

Principal ideals:

$$(x), \quad (y), \quad (xy), \quad (x^2), \quad (x^2 + y^2).$$

Two-generated ideals:

$$(x, y), \quad (x^2, xy), \quad (x, y^2), \quad (x^2, y).$$

An element of (x^2, xy) has the form

$$f(x, y)x^2 + g(x, y)xy.$$

So

$$(x^2, xy) = x(x, y).$$

6.6 6. Example: $k[x, y]/(xy)$

Let

$$A = k[x, y], \qquad M = A/(xy).$$

Meaning of the quotient. In this quotient,

$$\overline{x}\,\overline{y} = 0,$$

but neither $\overline{x}$ nor $\overline{y}$ is zero. So the quotient has zero divisors.

Geometry. The variety $V(xy)$ is the union of the coordinate axes:

$$V(xy) = V(x) \cup V(y).$$

So $k[x, y]/(xy)$ is the coordinate ring of two lines crossing at the origin.

Minimal primes. Since $(xy) \subseteq (x)$ and $(xy) \subseteq (y)$, and both (x) and (y) are prime,

$$\mathrm{Min}(A/(xy)) = \{(x), (y)\}.$$

Also,

$$\sqrt{(xy)} = (x) \cap (y).$$

Associated primes. Consider $\overline{x} \in A/(xy)$. Then

$$\mathrm{Ann}_A(\overline{x}) = \{f \in A : fx \in (xy)\}.$$

This happens exactly when f is divisible by y, so

$$\mathrm{Ann}_A(\overline{x}) = (y).$$

Similarly,

$$\mathrm{Ann}_A(\overline{y}) = (x).$$

Therefore

$$\mathrm{Ass}_A(A/(xy)) = \{(x), (y)\}.$$

So here

$$\mathrm{Ass}_A(A/(xy)) = \mathrm{Min}(A/(xy)).$$

Support.

$$\mathrm{Supp}_A(A/(xy)) = V(xy) = V(x) \cup V(y).$$

6.7 7. Example: $k[x, y]/(x^2)$

Let

$$A = k[x, y], \qquad M = A/(x^2).$$

Geometry. This is a nonreduced double line along $x = 0$.

Minimal primes. Since

$$\sqrt{(x^2)} = (x),$$

we have

$$\mathrm{Min}(A/(x^2)) = \{(x)\}.$$

Associated primes. The class $\overline{x}$ satisfies

$$x \cdot \overline{x} = 0,$$

and

$$\mathrm{Ann}_A(\overline{x}) = \{f : fx \in (x^2)\} = (x).$$

Therefore

$$\mathrm{Ass}_A(A/(x^2)) = \{(x)\}.$$

Again,

$$\mathrm{Ass}_A(A/(x^2)) = \mathrm{Min}(A/(x^2)).$$

6.8 8. Example: $k[x, y]/(x^2, xy)$

Let

$$A = k[x, y], \qquad I = (x^2, xy) = x(x, y), \qquad M = A/I.$$

This is a standard example with an embedded associated prime.

Understanding the ideal. Every element of I has the form

$$f(x, y)x^2 + g(x, y)xy.$$

Equivalently,

$$I = x(x, y).$$

Minimal primes. Since every generator of I is divisible by x, we have

$$I \subseteq (x).$$

Also,

$$\sqrt{I} = (x).$$

Hence

$$\mathrm{Min}(A/I) = \{(x)\}.$$

Associated prime (x). Consider $\overline{y} \in A/I$. Then

$$\mathrm{Ann}_A(\overline{y}) = \{f : fy \in (x^2, xy)\}.$$

This happens exactly when f is divisible by x. So

$$\mathrm{Ann}_A(\overline{y}) = (x).$$

Therefore

$$(x) \in \mathrm{Ass}_A(A/I).$$

Associated prime (x, y). Now consider $\overline{x} \in A/I$. Since

$$x^2 \in I, \qquad xy \in I,$$

we get

$$x \cdot \overline{x} = 0, \qquad y \cdot \overline{x} = 0.$$

So

$$(x, y) \subseteq \mathrm{Ann}_A(\overline{x}).$$

Conversely, if $f\overline{x} = 0$, then

$$fx \in (x^2, xy) = x(x, y),$$

hence

$$f \in (x, y).$$

Therefore

$$\mathrm{Ann}_A(\overline{x}) = (x, y).$$

So

$$(x, y) \in \mathrm{Ass}_A(A/I).$$

Thus

$$\mathrm{Ass}_A(A/(x^2, xy)) = \{(x), (x, y)\}.$$

This is the important phenomenon:

$$\mathrm{Min}(A/(x^2, xy)) = \{(x)\},$$

but

$$\mathrm{Ass}_A(A/(x^2, xy)) = \{(x), (x, y)\}.$$

So (x, y) is an *embedded associated prime*.

6.9 9. Example: $k[x,y]/(x,y)$

Let

$$A = k[x,y], \qquad M = A/(x,y).$$

Since both x and y become zero, we have

$$A/(x,y) \cong k.$$

As an A-module,

$$\mathrm{Ann}_A(M) = (x,y).$$

Since (x,y) is prime, it follows that

$$\mathrm{Ass}_A(A/(x,y)) = \{(x,y)\}, \qquad \mathrm{Min}(A/(x,y)) = \{(x,y)\}.$$

Geometrically, this is the origin.

6.10 10. Distinguished open sets $D(f)$ in $\mathrm{Spec}(k[x,y])$

The open set $D(x)$.

$$D(x) = \{\mathfrak{p} \in \mathrm{Spec}(k[x,y]) : x \notin \mathfrak{p}\}.$$

This is the complement of $V(x)$, so it is the part of the spectrum where x does not vanish.

The open set $D(y)$.

$$D(y) = \{\mathfrak{p} \in \mathrm{Spec}(k[x,y]) : y \notin \mathfrak{p}\}.$$

The open set $D(xy)$. Since

$$D(fg) = D(f) \cap D(g),$$

we get

$$D(xy) = D(x) \cap D(y).$$

Equivalently,

$$V(xy) = V(x) \cup V(y),$$

so $D(xy)$ is the complement of the union of the two coordinate axes.

6.11 11. Summary table

Let $A = k[x,y]$ unless otherwise stated.

Ring or module	Ann	Min	Ass
$k[x]$	(0)	$\{(0)\}$	$\{(0)\}$
$k[x]/(x^2)$	(x^2)	$\{(x)\}$	$\{(x)\}$
$k[x]/(x(x-1))$	$(x(x-1))$	$\{(x),(x-1)\}$	$\{(x),(x-1)\}$
$k[x,y]$	(0)	$\{(0)\}$	$\{(0)\}$
$k[x,y]/(xy)$	(xy)	$\{(x),(y)\}$	$\{(x),(y)\}$
$k[x,y]/(x^2)$	(x^2)	$\{(x)\}$	$\{(x)\}$
$k[x,y]/(x^2,xy)$	(x^2,xy)	$\{(x)\}$	$\{(x),(x,y)\}$
$k[x,y]/(x,y)$	(x,y)	$\{(x,y)\}$	$\{(x,y)\}$

6.12 12. Main relation between minimal and associated primes

For a finitely generated module M over a Noetherian ring A, one has

$$\mathrm{Min}(\mathrm{Supp}(M)) \subseteq \mathrm{Ass}_A(M) \subseteq \mathrm{Supp}_A(M).$$

For $M = A/I$, this becomes

$$\mathrm{Min}(A/I) \subseteq \mathrm{Ass}_A(A/I) \subseteq V(I).$$

So:

- every minimal prime is associated,
- but there may be extra associated primes,
- those extra ones are called embedded primes.

The quotient

$$k[x,y]/(x^2, xy)$$

is the standard example:

$$\mathrm{Min} = \{(x)\}, \qquad \mathrm{Ass} = \{(x), (x,y)\}.$$

6.13 13. Very concrete intuition

For $k[x,y]/(xy)$:

$$\overline{x} \neq 0, \qquad \overline{y} \neq 0, \qquad \overline{x}\,\overline{y} = 0.$$

So x and y kill each other:

$$\mathrm{Ann}(\overline{x}) = (y), \qquad \mathrm{Ann}(\overline{y}) = (x).$$

For $k[x,y]/(x^2)$:

$$\overline{x} \neq 0, \qquad \overline{x}^2 = 0.$$

So $\overline{x}$ is nilpotent, and

$$\mathrm{Ann}(\overline{x}) = (x).$$

For $k[x,y]/(x^2, xy)$:

$$x^2 = 0, \qquad xy = 0$$

in the quotient, so $\overline{x}$ is killed by both x and y:

$$\mathrm{Ann}(\overline{x}) = (x,y).$$

But $\overline{y}$ is killed only by multiples of x:

$$\mathrm{Ann}(\overline{y}) = (x).$$

That is why both

$$(x) \quad \text{and} \quad (x,y)$$

appear in Ass.

7 Primary decomposition, radicals, and explicit residue classes in quotient rings

Convention. Throughout, k is a field, all rings are commutative with 1, and all modules are unital. We continue the discussion for the rings

$$k[x], \qquad k[x,y].$$

7.1 1. Radicals

Definition. If $I \subseteq A$ is an ideal, its radical is

$$\sqrt{I} = \{a \in A : \exists n \geq 1 \text{ such that } a^n \in I\}.$$

Meaning. An element lies in $\sqrt{I}$ if some power of it already lies in I.

Examples.

$$\sqrt{(x^2)} = (x), \qquad \sqrt{(xy)} = (x) \cap (y), \qquad \sqrt{(x^2, xy)} = (x).$$

Why $\sqrt{(x^2)} = (x)$? If $f^n \in (x^2)$, then in particular $x \mid f^n$, hence $x \mid f$, so $f \in (x)$. Conversely, if $f \in (x)$, then $f = xg$, so

$$f^2 = x^2 g^2 \in (x^2),$$

hence $f \in \sqrt{(x^2)}$.

Why $\sqrt{(xy)} = (x) \cap (y)$? In $k[x,y]$, the ideals (x) and (y) are prime, and

$$(xy) \subseteq (x) \cap (y).$$

Also,

$$(x) \cap (y) = (xy)$$

in the UFD $k[x,y]$, because a polynomial divisible by both x and y is divisible by xy. Since (xy) is already radical, we get

$$\sqrt{(xy)} = (xy) = (x) \cap (y).$$

Why $\sqrt{(x^2, xy)} = (x)$? We have

$$(x^2, xy) = x(x,y) \subseteq (x),$$

so $\sqrt{(x^2, xy)} \subseteq (x)$. Conversely, $x^2 \in (x^2, xy)$, so $x \in \sqrt{(x^2, xy)}$, hence every multiple of x lies in the radical. Thus

$$\sqrt{(x^2, xy)} = (x).$$

7.2 2. Primary ideals

Definition. An ideal $Q \subseteq A$ is called *primary* if whenever

$$ab \in Q,$$

then either

$$a \in Q$$

or

$$b^n \in Q \quad \text{for some } n \geq 1.$$

Equivalent viewpoint. Q is primary if in A/Q, every zero divisor is nilpotent.

Associated prime of a primary ideal. If Q is primary, then $\sqrt{Q}$ is a prime ideal. In that case one says that Q is $\mathfrak{p}$-primary, where

$$\mathfrak{p} = \sqrt{Q}.$$

Examples.

- $(x^2) \subset k[x]$ is (x)-primary.
- $(x^2) \subset k[x,y]$ is (x)-primary.
- $(x,y) \subset k[x,y]$ is prime, hence also (x,y)-primary.
- $(x) \subset k[x,y]$ is prime, hence also (x)-primary.
- $(xy) \subset k[x,y]$ is *not* primary, because

$$xy \in (xy), \qquad x \notin (xy), \qquad y^n \notin (xy) \ \forall n \geq 1.$$

7.3 3. Primary decomposition

Definition. A *primary decomposition* of an ideal I is an expression

$$I = Q_1 \cap \cdots \cap Q_r$$

where each Q_i is primary.

If the radicals

$$\sqrt{Q_1}, \ldots, \sqrt{Q_r}$$

are distinct and no Q_i can be omitted, one calls it an irredundant primary decomposition.

Important fact in Noetherian rings. Every ideal in a Noetherian ring admits a primary decomposition.

Since $k[x]$ and $k[x,y]$ are Noetherian, this applies to all our examples.

7.4 4. Explicit residue classes: general principle

If $A = R/I$, then elements of A are equivalence classes

$$f + I, \qquad f \in R.$$

We usually write $\overline{f}$ for the class of f.

Two polynomials $f, g \in R$ define the same class iff

$$f - g \in I.$$

So to understand R/I, we want a simple description of representatives of classes.

7.5 5. Example: $k[x]/(x^2)$

Let

$$A = k[x]/(x^2).$$

Primary decomposition. The ideal (x^2) is already primary:

$$(x^2) \text{ is } (x)\text{-primary}.$$

So the primary decomposition is just

$$(x^2).$$

Radical.

$$\sqrt{(x^2)} = (x).$$

Explicit residue classes. Every polynomial

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots$$

is congruent modulo (x^2) to

$$a_0 + a_1 x.$$

So every class has a unique representative of the form

$$a + bx.$$

Thus

$$k[x]/(x^2) = \{a + b\overline{x} : a, b \in k,\ \overline{x}^2 = 0\}.$$

Multiplication rule.

$$(a + b\overline{x})(c + d\overline{x}) = ac + (ad + bc)\overline{x},$$

because $\overline{x}^2 = 0$.

Interpretation. This is the ring of dual numbers over k.

7.6 6. Example: $k[x]/(x(x-1))$

Let

$$A = k[x]/(x(x-1)).$$

Primary decomposition. Since (x) and $(x-1)$ are distinct maximal ideals in $k[x]$,

$$(x(x-1)) = (x) \cap (x-1).$$

Each factor is prime, hence primary. So this is a primary decomposition.

Radical. Because $(x) \cap (x-1)$ is already radical,

$$\sqrt{(x(x-1))} = (x) \cap (x-1).$$

Explicit residue classes. Modulo $x(x-1)$, every polynomial class can be represented by a polynomial of degree at most 1:

$$a + bx.$$

Indeed, division by the degree-2 polynomial $x(x-1)$ gives such a remainder.

So every class has a unique representative

$$a + b\overline{x}.$$

Another description via Chinese remainder theorem.

$$k[x]/(x(x-1)) \cong k[x]/(x) \times k[x]/(x-1) \cong k \times k.$$

Under this isomorphism,

$$f(x) \longmapsto (f(0), f(1)).$$

So classes can also be described as pairs

$$(a, b) \in k \times k.$$

How does $a + bx$ correspond to a pair?

$$a + bx \longmapsto (a, a+b).$$

Special classes.

$$\overline{x} \longmapsto (0, 1), \qquad \overline{x-1} \longmapsto (-1, 0).$$

7.7 7. Example: $k[x, y]/(xy)$

Let

$$A = k[x, y]/(xy).$$

Primary decomposition. Since (x) and (y) are prime ideals and

$$(x) \cap (y) = (xy),$$

we get the primary decomposition

$$(xy) = (x) \cap (y).$$

Radical. Because $(xy) = (x) \cap (y)$ is radical,

$$\sqrt{(xy)} = (xy) = (x) \cap (y).$$

Explicit residue classes. Every polynomial can be written as

$$f(x, y) = a + xg(x) + yh(y) + xy\,q(x, y),$$

for suitable polynomials g, h, q. Modulo (xy), the last term vanishes, so every class can be written as

$$a + xg(x) + yh(y).$$

Equivalently, every class has a representative of the form

$$a + p(x) + q(y),$$

where $p(0) = 0$ and $q(0) = 0$.

So the quotient can be described as

$$k[x, y]/(xy) \cong \{a + p(x) + q(y) : a \in k,\ p \in xk[x],\ q \in yk[y]\}.$$

Why this is natural. All mixed terms

$$x^i y^j \quad (i, j \geq 1)$$

vanish in the quotient, because each is divisible by xy.

Thus only:

- the constant term,
- pure x-terms,
- pure y-terms

survive.

Examples of classes.

$$\overline{x^2 + y^3 + 5} = 5 + \overline{x^2} + \overline{y^3},$$

but

$$\overline{x^2 y + y^4 + 7} = 7 + \overline{y^4},$$

because $x^2 y \in (xy)$.

Multiplication. Products of pure x-terms with pure y-terms are zero:

$$\overline{x^m}\,\overline{y^n} = 0 \qquad (m, n \geq 1).$$

7.8 8. Example: $k[x, y]/(x^2)$

Let

$$A = k[x, y]/(x^2).$$

Primary decomposition. The ideal (x^2) is already (x)-primary. So its primary decomposition is simply

$$(x^2).$$

Radical.

$$\sqrt{(x^2)} = (x).$$

Explicit residue classes. Every polynomial can be written uniquely in the form

$$f(x,y) = g_0(y) + xg_1(y) + x^2 g_2(x,y).$$

Modulo (x^2), the last term vanishes, so every class has a unique representative

$$g_0(y) + xg_1(y),$$

where $g_0(y), g_1(y) \in k[y]$.

Thus

$$k[x,y]/(x^2) \cong k[y] \oplus xk[y]$$

as a $k[y]$-module, with the relation

$$\overline{x}^2 = 0.$$

Examples.

$$\overline{y^3 + xy + 7} = y^3 + xy + 7,$$

while

$$\overline{x^3 + x^2y + y^2} = y^2,$$

because $x^3, x^2y \in (x^2)$.

Multiplication rule. If

$$a(y) + xb(y), \qquad c(y) + xd(y)$$

are two classes, then

$$(a(y) + xb(y))(c(y) + xd(y)) = a(y)c(y) + x(a(y)d(y) + b(y)c(y)),$$

because $x^2 = 0$.

7.9 9. Example: $k[x,y]/(x^2, xy)$

Let

$$A = k[x,y]/(x^2, xy).$$

First note. The relations in the quotient are

$$x^2 = 0, \qquad xy = 0.$$

Primary decomposition. A standard decomposition is

$$(x^2, xy) = (x) \cap (x^2, y).$$

We check it.

First, if $f \in (x) \cap (x^2, y)$, then

$$f = xg$$

for some g, and also

$$f = x^2a + yb$$

for some a, b. Since f is divisible by x, the term yb must also be divisible by x, hence $b \in (x)$, say $b = xc$. Thus

$$f = x^2a + y(xc) = x^2a + xyc \in (x^2, xy).$$

So

$$(x) \cap (x^2, y) \subseteq (x^2, xy).$$

The reverse inclusion is obvious, since both x^2 and xy lie in both ideals. Hence

$$(x^2, xy) = (x) \cap (x^2, y).$$

Why is this a primary decomposition?

- (x) is prime, hence (x)-primary.
- (x^2, y) is (x, y)-primary, because

$$\sqrt{(x^2, y)} = (x, y).$$

So this is a primary decomposition.

Radical. Since

$$(x^2, xy) = x(x, y),$$

all elements are multiples of x, and $x^2 \in (x^2, xy)$, hence

$$\sqrt{(x^2, xy)} = (x).$$

Explicit residue classes. Because

$$x^2 = 0, \qquad xy = 0,$$

every monomial containing x^2 vanishes, and every monomial containing both x and y also vanishes.

So only:

- pure y-powers,
- the linear x-term

survive.

Thus every class has a unique representative of the form

$$g(y) + ax,$$

where $g(y) \in k[y]$ and $a \in k$.

Indeed, higher terms $xy, xy^2, \dots$ vanish, and $x^2, x^2y, \dots$ vanish too.

So

$$k[x,y]/(x^2, xy) \cong k[y] \oplus kx$$

as a k-vector space, with

$$x^2 = 0, \qquad xy = 0.$$

Examples.

$$\overline{y^4 + 3x + 7} = y^4 + 3x + 7,$$

but

$$\overline{x^2 + xy + y^2} = y^2.$$

Multiplication rule. If

$$g(y) + ax, \qquad h(y) + bx$$

are two classes, then

$$(g(y) + ax)(h(y) + bx) = g(y)h(y) + a\,x\,h(0) + b\,x\,g(0),$$

because

$$x \cdot y = 0, \qquad x^2 = 0.$$

Equivalently, only the constant terms of $g(y)$ and $h(y)$ interact with x.

7.10 10. Example: $k[x,y]/(x,y)$

Let

$$A = k[x,y]/(x,y).$$

Primary decomposition. Since (x,y) is maximal, it is prime and therefore primary. So the primary decomposition is simply

$$(x,y).$$

Radical.

$$\sqrt{(x,y)} = (x,y).$$

Explicit residue classes. Both x and y vanish, so every polynomial is congruent to its constant term:

$$f(x,y) \equiv f(0,0) \pmod{(x,y)}.$$

Thus

$$k[x,y]/(x,y) \cong k.$$

7.11 11. Summary table

Ideal I	$\sqrt{I}$	Primary decomposition of I	Typical class in A/I
$(x^2) \subset k[x]$	(x)	(x^2)	$a + bx$
$(x(x-1)) \subset k[x]$	$(x) \cap (x-1)$	$(x) \cap (x-1)$	$a + bx$
$(xy) \subset k[x,y]$	$(x) \cap (y)$	$(x) \cap (y)$	$a + p(x) + q(y)$
$(x^2) \subset k[x,y]$	(x)	(x^2)	$g_0(y) + xg_1(y)$
$(x^2, xy) \subset k[x,y]$	(x)	$(x) \cap (x^2, y)$	$g(y) + ax$
$(x, y) \subset k[x,y]$	(x, y)	(x, y)	a

7.12 12. Geometric meaning

$k[x]/(x^2)$. This is a doubled point at $x = 0$: same underlying point, but with nilpotent structure.

$k[x]/(x(x-1))$. This is two reduced points, at 0 and 1.

$k[x,y]/(xy)$. This is the union of the two coordinate axes:

$$V(x) \cup V(y).$$

$k[x,y]/(x^2)$. This is a doubled line along $x = 0$.

$k[x,y]/(x^2, xy)$. This has underlying set $x = 0$, but with extra embedded structure at the origin.

$k[x,y]/(x,y)$. This is the origin.

7.13 13. Final conceptual summary

- The *radical* $\sqrt{I}$ tells us the underlying geometric set.
- The *primary decomposition* separates an ideal into pieces corresponding to prime components and possible embedded components.
- The *explicit residue classes* tell us concretely what elements survive in the quotient ring.

The three key model examples are:

$$k[x,y]/(xy), \qquad k[x,y]/(x^2), \qquad k[x,y]/(x^2, xy).$$

They illustrate three different situations:

- reduced reducible structure: (xy),
- nonreduced irreducible structure: (x^2),
- nonreduced structure with an embedded component: (x^2, xy).

8 Explicit computations of $\mathrm{Ann}_A(\overline{f})$, $\mathrm{Ass}_A(A/I)$, and $\mathrm{Min}(A/I)$

Convention. In this section A denotes the ambient polynomial ring, $I \subseteq A$ an ideal, and $\overline{f}$ the class of $f \in A$ in the quotient A/I.

8.1 1. General formula: annihilator as a colon ideal

For any $f \in A$, the annihilator of $\overline{f} \in A/I$ is

$$\mathrm{Ann}_A(\overline{f}) = \{a \in A : a\overline{f} = 0\} = \{a \in A : af \in I\} = (I : f).$$

So in practice one computes

$$\mathrm{Ann}_A(\overline{f}) = (I : f).$$

This is the most useful concrete rule.

Important remarks.

- $\mathrm{Ann}_A(\overline{1}) = I$.
- Associated primes of A/I are precisely the prime ideals that occur as annihilators of some element $\overline{f} \in A/I$.
- Minimal primes of A/I are the minimal prime ideals containing I, equivalently the minimal primes of $\sqrt{I}$.

8.2 2. The quotient $k[x]/(x^2)$

Let

$$A = k[x], \qquad I = (x^2), \qquad M = A/I.$$

We know

$$\sqrt{I} = (x), \qquad \mathrm{Min}(A/I) = \{(x)\}.$$

Computation 1: $\mathrm{Ann}_A(\overline{1})$**.**

$$\mathrm{Ann}_A(\overline{1}) = (I : 1) = I = (x^2).$$

Computation 2: $\mathrm{Ann}_A(\overline{x})$**.**

$$\mathrm{Ann}_A(\overline{x}) = (x^2 : x).$$

Now

$$g(x) \in (x^2 : x) \iff g(x)x \in (x^2) \iff x \mid g(x).$$

Hence

$$\mathrm{Ann}_A(\overline{x}) = (x).$$

Computation 3: $\mathrm{Ann}_A(\overline{1+x})$**.** Since $1 + x$ is relatively prime to x^2, if

$$g(x)(1 + x) \in (x^2),$$

then necessarily $g(x) \in (x^2)$. Therefore
$$\mathrm{Ann}_A(\overline{1+x}) = (x^2).$$

Associated primes. The annihilator (x) is prime and occurs for the element $\overline{x}$, so
$$\mathrm{Ass}_A(A/I) = \{(x)\}.$$

Thus
$$\mathrm{Ass}_A(A/(x^2)) = \mathrm{Min}(A/(x^2)) = \{(x)\}.$$

8.3 3. The quotient $k[x]/(x(x-1))$

Let
$$A = k[x], \qquad I = (x(x-1)), \qquad M = A/I.$$

We know
$$I = (x) \cap (x-1), \qquad \mathrm{Min}(A/I) = \{(x), (x-1)\}.$$

Computation 1: $\mathrm{Ann}_A(\overline{1})$**.**
$$\mathrm{Ann}_A(\overline{1}) = I = (x(x-1)).$$

Computation 2: $\mathrm{Ann}_A(\overline{x})$**.**
$$\mathrm{Ann}_A(\overline{x}) = (x(x-1) : x).$$

Now
$$g(x) \in (x(x-1) : x) \iff g(x)x \in (x(x-1)) \iff x-1 \mid g(x).$$
So
$$\mathrm{Ann}_A(\overline{x}) = (x-1).$$

Computation 3: $\mathrm{Ann}_A(\overline{x-1})$**.** Similarly,
$$\mathrm{Ann}_A(\overline{x-1}) = (x(x-1) : x-1) = (x).$$

A convenient global description. Under the Chinese remainder isomorphism
$$k[x]/(x(x-1)) \cong k \times k, \qquad \overline{f} \longmapsto (f(0), f(1)),$$
the annihilator depends on which coordinates vanish:
$$\mathrm{Ann}_A(\overline{f}) = \begin{cases} (x(x-1)), & f(0) \neq 0,\ f(1) \neq 0, \\ (x), & f(0) = 0,\ f(1) \neq 0, \\ (x-1), & f(0) \neq 0,\ f(1) = 0, \\ A, & f(0) = f(1) = 0. \end{cases}$$

Associated primes. Since (x) and $(x-1)$ occur as annihilators, we get
$$\mathrm{Ass}_A(A/I) = \{(x), (x-1)\}.$$

Thus
$$\mathrm{Ass}_A(A/(x(x-1))) = \mathrm{Min}(A/(x(x-1))) = \{(x), (x-1)\}.$$

8.4 4. The quotient $k[x,y]/(xy)$

Let
$$A = k[x,y], \qquad I = (xy), \qquad M = A/I.$$

We know
$$I = (x) \cap (y), \qquad \sqrt{I} = (xy) = (x) \cap (y), \qquad \mathrm{Min}(A/I) = \{(x),(y)\}.$$

Computation 1: $\mathrm{Ann}_A(\overline{1})$**.**
$$\mathrm{Ann}_A(\overline{1}) = I = (xy).$$

Computation 2: $\mathrm{Ann}_A(\overline{x})$**.**
$$\mathrm{Ann}_A(\overline{x}) = (xy : x).$$

Now
$$g(x,y) \in (xy : x) \iff g(x,y)x \in (xy) \iff y \mid g(x,y).$$

Hence
$$\mathrm{Ann}_A(\overline{x}) = (y).$$

Computation 3: $\mathrm{Ann}_A(\overline{y})$**.** Similarly,
$$\mathrm{Ann}_A(\overline{y}) = (xy : y) = (x).$$

Computation 4: higher pure powers. For every $m \geq 1$,
$$\mathrm{Ann}_A(\overline{x^m}) = (y), \qquad \mathrm{Ann}_A(\overline{y^m}) = (x).$$

Indeed,
$$gx^m \in (xy) \iff y \mid g, \qquad gy^m \in (xy) \iff x \mid g.$$

Associated primes. Since (x) and (y) occur as annihilators,
$$\mathrm{Ass}_A(A/I) = \{(x),(y)\}.$$

Thus
$$\mathrm{Ass}_A(A/(xy)) = \mathrm{Min}(A/(xy)) = \{(x),(y)\}.$$

8.5 5. The quotient $k[x,y]/(x^2)$

Let
$$A = k[x,y], \qquad I = (x^2), \qquad M = A/I.$$

We know
$$\sqrt{I} = (x), \qquad \mathrm{Min}(A/I) = \{(x)\}.$$

Computation 1: $\mathrm{Ann}_A(\overline{1})$**.**
$$\mathrm{Ann}_A(\overline{1}) = I = (x^2).$$

Computation 2: $\mathrm{Ann}_A(\overline{x})$**.**

$$\mathrm{Ann}_A(\overline{x}) = (x^2 : x) = (x).$$

Computation 3: $\mathrm{Ann}_A(\overline{y})$**.**

$$\mathrm{Ann}_A(\overline{y}) = (x^2 : y).$$

Now

$$g(x,y) \in (x^2 : y) \iff g(x,y)y \in (x^2).$$

Since y is relatively prime to x^2, this happens iff

$$g(x,y) \in (x^2).$$

So

$$\mathrm{Ann}_A(\overline{y}) = (x^2).$$

Computation 4: elements of the form $\overline{xg(y)}$**.** If $g(y) \neq 0$, then

$$\mathrm{Ann}_A(\overline{xg(y)}) = (x).$$

Indeed,

$$h(x,y) \cdot xg(y) \in (x^2) \iff x \mid h(x,y).$$

Associated primes. The prime ideal (x) occurs as $\mathrm{Ann}_A(\overline{x})$, hence

$$\mathrm{Ass}_A(A/I) = \{(x)\}.$$

Thus

$$\mathrm{Ass}_A(A/(x^2)) = \mathrm{Min}(A/(x^2)) = \{(x)\}.$$

8.6 6. The quotient $k[x,y]/(x^2, xy)$

Let

$$A = k[x,y], \qquad I = (x^2, xy), \qquad M = A/I.$$

We know

$$I = (x) \cap (x^2, y), \qquad \sqrt{I} = (x), \qquad \mathrm{Min}(A/I) = \{(x)\}.$$

Also

$$\mathrm{Ass}_A(A/I) = \{(x), (x,y)\},$$

and now we compute these explicitly from concrete elements.

Normal form of elements. Every class in A/I has a unique representative of the form

$$g(y) + ax, \qquad g(y) \in k[y],\ a \in k,$$

because

$$x^2 = 0, \qquad xy = 0$$

in the quotient.

Computation 1: $\mathrm{Ann}_A(\overline{1})$**.**

$$\mathrm{Ann}_A(\overline{1}) = I = (x^2, xy).$$

Computation 2: $\mathrm{Ann}_A(\overline{x})$**.**

$$\mathrm{Ann}_A(\overline{x}) = (x^2, xy : x).$$

Now

$$g(x,y) \in (x^2, xy : x) \iff g(x,y)x \in (x^2, xy) = x(x,y) \iff g(x,y) \in (x,y).$$

So

$$\mathrm{Ann}_A(\overline{x}) = (x,y).$$

Computation 3: $\mathrm{Ann}_A(\overline{y})$**.**

$$\mathrm{Ann}_A(\overline{y}) = (x^2, xy : y).$$

Now

$$g(x,y)y \in (x^2, xy)$$

forces $g(x,y)$ to be divisible by x, and every multiple of x works because

$$x \cdot y = xy \in I.$$

Hence

$$\mathrm{Ann}_A(\overline{y}) = (x).$$

Computation 4: higher powers of y**.** For every $m \geq 1$,

$$\mathrm{Ann}_A(\overline{y^m}) = (x).$$

Indeed,

$$g(x,y)y^m \in (x^2, xy) \iff x \mid g(x,y).$$

Computation 5: classification by the normal form $g(y) + ax$**.**

Let

$$z = \overline{g(y) + ax} \in A/I.$$

Then there are four essentially different cases.

Case 1: $z = 0$. Then

$$\mathrm{Ann}_A(z) = A.$$

Case 2: $g(y) = 0$ *and* $a \neq 0$*, so* $z = \overline{ax}$. Then

$$\mathrm{Ann}_A(z) = (x,y).$$

Case 3: $g(y) \neq 0$ *and* $g(0) = 0$. Then $g(y) \in yk[y]$, and

$$\mathrm{Ann}_A(z) = (x).$$

Indeed, x kills every such element because

$$xg(y) = 0, \qquad x \cdot ax = 0$$

in the quotient, and nothing outside (x) can kill a nonzero y-polynomial part.

Case 4: $g(0) \neq 0$. Then

$$\mathrm{Ann}_A(z) = I.$$

Indeed, such an element has a nonzero constant term in its $k[y]$-part, so no nonzero class in A/I kills it.

Associated primes. From the explicit annihilators we see:

$$\mathrm{Ann}_A(\overline{y}) = (x), \qquad \mathrm{Ann}_A(\overline{x}) = (x, y).$$

Therefore

$$\mathrm{Ass}_A(A/I) = \{(x), (x, y)\}.$$

This is the standard example where

$$\mathrm{Min}(A/I) = \{(x)\}$$

but

$$\mathrm{Ass}_A(A/I) = \{(x), (x, y)\}.$$

So (x, y) is an embedded associated prime.

8.7 7. The quotient $k[x, y]/(x, y)$

Let

$$A = k[x, y], \qquad I = (x, y), \qquad M = A/I.$$

We know

$$A/I \cong k, \qquad \mathrm{Min}(A/I) = \{(x, y)\}.$$

Computation 1: $\mathrm{Ann}_A(\overline{1})$.

$$\mathrm{Ann}_A(\overline{1}) = I = (x, y).$$

Computation 2: any nonzero class. Every nonzero class in $A/I \cong k$ is of the form $\overline{c}$ with $c \in k^\times$, and

$$\mathrm{Ann}_A(\overline{c}) = (x, y),$$

because multiplying by a nonzero scalar does not change the annihilator.

Computation 3: the zero class.

$$\mathrm{Ann}_A(\overline{0}) = A.$$

Associated primes. The only prime annihilator that occurs for a nonzero element is

$$(x, y).$$

Hence

$$\mathrm{Ass}_A(A/I) = \{(x, y)\}.$$

Thus

$$\mathrm{Ass}_A(A/(x, y)) = \mathrm{Min}(A/(x, y)) = \{(x, y)\}.$$

8.8 8. Summary table of explicit annihilators

A/I	element $\overline{f}$	$\mathrm{Ann}_A(\overline{f})$
$k[x]/(x^2)$	$\overline{1}$	(x^2)
$k[x]/(x^2)$	$\overline{x}$	(x)
$k[x]/(x(x-1))$	$\overline{x}$	$(x-1)$
$k[x]/(x(x-1))$	$\overline{x-1}$	(x)
$k[x,y]/(xy)$	$\overline{x}$	(y)
$k[x,y]/(xy)$	$\overline{y}$	(x)
$k[x,y]/(x^2)$	$\overline{x}$	(x)
$k[x,y]/(x^2)$	$\overline{y}$	(x^2)
$k[x,y]/(x^2,xy)$	$\overline{x}$	(x,y)
$k[x,y]/(x^2,xy)$	$\overline{y}$	(x)
$k[x,y]/(x,y)$	$\overline{1}$	(x,y)

8.9 9. Summary table of Ass and Min

A/I	$\mathrm{Ass}_A(A/I)$	$\mathrm{Min}(A/I)$
$k[x]/(x^2)$	$\{(x)\}$	$\{(x)\}$
$k[x]/(x(x-1))$	$\{(x),(x-1)\}$	$\{(x),(x-1)\}$
$k[x,y]/(xy)$	$\{(x),(y)\}$	$\{(x),(y)\}$
$k[x,y]/(x^2)$	$\{(x)\}$	$\{(x)\}$
$k[x,y]/(x^2,xy)$	$\{(x),(x,y)\}$	$\{(x)\}$
$k[x,y]/(x,y)$	$\{(x,y)\}$	$\{(x,y)\}$

8.10 10. Final interpretation

These explicit computations show the following pattern:

- $\mathrm{Min}(A/I)$ describes the geometric irreducible components of the underlying set.
- $\mathrm{Ass}_A(A/I)$ detects not only minimal components, but also possible embedded components.
- The concrete annihilators $\mathrm{Ann}_A(\overline{f})$ are the bridge between the module A/I and the prime ideals that appear in $\mathrm{Ass}_A(A/I)$.

The most important example remains

$$k[x,y]/(x^2,xy),$$

because it shows explicitly how the two annihilators

$$\mathrm{Ann}_A(\overline{y})=(x), \qquad \mathrm{Ann}_A(\overline{x})=(x,y)$$

produce

$$\mathrm{Ass}_A(A/(x^2,xy))=\{(x),(x,y)\},$$

while the only minimal prime is

$$\mathrm{Min}(A/(x^2,xy))=\{(x)\}.$$

9 Local rings, domination, valuation rings, value groups, and valuations

Convention. Throughout, all rings are commutative with 1. If A is a domain, its field of fractions is denoted $\mathrm{Frac}(A)$. If K is a field and $A \subseteq K$ is a subring, we view A as a domain inside K.

9.1 1. Local rings

Definition. A ring A is called *local* if it has exactly one maximal ideal. We usually denote it by

$$\mathfrak{m},$$

and then write the local ring as

$$(A, \mathfrak{m}).$$

Equivalent characterization. A ring A is local if and only if the set of nonunits of A is an ideal. That ideal is then the unique maximal ideal $\mathfrak{m}$. Thus

$$A^\times = A \setminus \mathfrak{m}.$$

So in a local ring every element is either:

- a unit, or
- an element of the unique maximal ideal.

Example 1: a field. Every field K is local. Its only maximal ideal is

$$(0),$$

because every nonzero element is invertible.

Example 2: $k[x]_{(x)}$. Let

$$A = k[x]_{(x)}.$$

This is the localization of $k[x]$ at the prime ideal (x). Its elements are fractions

$$\frac{f(x)}{g(x)} \qquad \text{with } g(0) \neq 0.$$

It is a local ring with maximal ideal

$$\mathfrak{m} = (x)k[x]_{(x)}.$$

An element $\frac{f}{g}$ is a unit if and only if $f(0) \neq 0$. So the nonunits are exactly those fractions divisible by x.

Example 3: $k[[t]]$. The formal power series ring

$$k[[t]]$$

is local with maximal ideal

$$(t).$$

Indeed, a power series

$$a_0 + a_1 t + a_2 t^2 + \cdots$$

is a unit if and only if $a_0 \neq 0$. Thus the nonunits are exactly the series with zero constant term.

Example 4: $\mathbb{Z}_{(p)}$. The localization

$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z},\ p \nmid b \right\}$$

is local with maximal ideal

$$p\mathbb{Z}_{(p)}.$$

A fraction $\frac{a}{b}$ is a unit if and only if $p \nmid a$.

9.2 2. Geometric meaning of local rings

A local ring is the algebraic model of looking at a space near one point.

For example:

$$k[x, y]_{(x,y)}$$

means: study the affine plane near the origin. More generally, for an algebraic variety X and a point $p \in X$, the local ring

$$\mathcal{O}_{X,p}$$

contains functions that are regular near p.

So global rings describe the whole space, while local rings describe behavior near one chosen point.

9.3 3. Domination of local rings

Let $(A, \mathfrak{m})$ and $(B, \mathfrak{n})$ be local rings.

Definition. We say that B *dominates* A if:

- $A \subseteq B$, and
- $\mathfrak{m} = A \cap \mathfrak{n}$.

Equivalently, one may say that the maximal ideal of A is the contraction of the maximal ideal of B. Informally: B is a larger local ring, but it keeps the same notion of which elements of A are nonunits.

Example.

$$k[x]_{(x)} \subseteq k[[x]]$$

and $k[[x]]$ dominates $k[x]_{(x)}$. Indeed:

- both are local,

- both have maximal ideal generated by x,
- and

$$(x)k[[x]] \cap k[x]_{(x)} = (x)k[x]_{(x)}.$$

9.4 4. Valuation rings

Let K be a field and let $A \subseteq K$ be a subring.

Definition. The ring A is called a *valuation ring* of K if for every nonzero $x \in K$,

$$x \in A \quad \text{or} \quad x^{-1} \in A.$$

This means that for every element of the field, the ring decides that either the element itself is regular, or its inverse is regular.

Intuition. A valuation ring measures whether an element has "nonnegative order" with respect to some notion of size, divisibility, or vanishing.

9.5 5. Basic examples of valuation rings

Example 1: $k[[t]] \subset k((t))$**.** Let

$$K = k((t)), \qquad A = k[[t]].$$

Every nonzero Laurent series can be written uniquely as

$$x = t^n u(t),$$

where $n \in \mathbb{Z}$ and $u(t) \in k[[t]]^\times$ is a unit.

If $n \geq 0$, then $x \in k[[t]]$. If $n < 0$, then

$$x^{-1} = t^{-n} u(t)^{-1} \in k[[t]].$$

So $k[[t]]$ is a valuation ring of $k((t))$.

Its maximal ideal is

$$(t),$$

and this is the standard example of a discrete valuation ring.

Example 2: $\mathbb{Z}_{(p)} \subset \mathbb{Q}$**.** For any nonzero rational number $x \in \mathbb{Q}$, either x has no extra factor of p in the denominator, in which case $x \in \mathbb{Z}_{(p)}$, or else $x^{-1} \in \mathbb{Z}_{(p)}$. So

$$\mathbb{Z}_{(p)}$$

is a valuation ring of $\mathbb{Q}$.

Example 3: $k[x]_{(x)} \subset k(x)$**.** Let

$$A = k[x]_{(x)} \subset k(x).$$

A rational function either has no pole at $x = 0$, in which case it lies in A, or else its inverse has no pole at 0. Hence A is a valuation ring of $k(x)$.

9.6 6. A local ring which is not a valuation ring

The ring

$$k[x,y]_{(x,y)}$$

is local, but it is *not* a valuation ring of $k(x,y)$.

Indeed, consider

$$\frac{x}{y} \in k(x,y).$$

Then neither $\frac{x}{y}$ nor $\frac{y}{x}$ lies in $k[x,y]_{(x,y)}$, since each has a pole along one coordinate axis.

So:

- every valuation ring is local,
- but not every local ring is a valuation ring.

9.7 7. Why valuation rings are local

Let $A \subseteq K$ be a valuation ring. Define

$$\mathfrak{m} = \{a \in A : a^{-1} \notin A\}.$$

This is exactly the set of nonunits of A. One checks that $\mathfrak{m}$ is an ideal, hence the unique maximal ideal of A. Therefore every valuation ring is a local ring.

9.8 8. Equivalent characterizations of valuation rings

Let A be a domain with fraction field $K = \mathrm{Frac}(A)$. Then the following are equivalent.

(1) Field comparability. For every $x \in K^\times$,

$$x \in A \quad \text{or} \quad x^{-1} \in A.$$

(2) All ideals are totally ordered by inclusion. For any two ideals $I, J \subseteq A$,

$$I \subseteq J \quad \text{or} \quad J \subseteq I.$$

(3) All principal ideals are totally ordered. For all $a, b \in A$,

$$(a) \subseteq (b) \quad \text{or} \quad (b) \subseteq (a).$$

(4) Divisibility is total. For all $a, b \in A$, either

$$a \mid b \quad \text{or} \quad b \mid a.$$

This shows that valuation rings are precisely rings where divisibility is completely comparable.

9.9 9. Discrete valuation rings

A *discrete valuation ring* (DVR) is a local PID with nonzero maximal ideal. Equivalently, it is a valuation ring whose value group is isomorphic to $\mathbb{Z}$.

Examples.

$$k[[t]], \qquad \mathbb{Z}_{(p)}, \qquad k[x]_{(x)}.$$

In a DVR every nonzero element can be written in the form

$$u\pi^n,$$

where

- u is a unit,
- π is a generator of the maximal ideal (a *uniformizer*),
- $n \geq 0$ if the element lies in the ring.

Examples of uniformizers:

$$t \text{ in } k[[t]], \qquad p \text{ in } \mathbb{Z}_{(p)}, \qquad x \text{ in } k[x]_{(x)}.$$

9.10 10. Maximal local subrings of a field

Let K be a field, and let Σ be the set of all local subrings of K, ordered by domination.

A fundamental fact is:

> A local subring $A \subseteq K$ is maximal with respect to domination if and only if A is a valuation ring of K.

Idea of the forward implication. Suppose A is local and maximal for domination, but not a valuation ring. Then there exists $x \in K^\times$ such that neither

$$x \in A \quad \text{nor} \quad x^{-1} \in A.$$

One can adjoin one of these elements and localize appropriately to obtain a strictly larger local subring of K dominating A, contradicting maximality. So a maximal local subring must be a valuation ring.

Idea of the converse. If A is a valuation ring, then its total comparability property is so strong that one cannot enlarge it inside K while preserving domination. Thus valuation rings are maximal local subrings of the field.

9.11 11. Units and the value group

Let A be a valuation ring of a field K. Let

$$U = A^\times$$

be the group of units of A. Since $U \subseteq K^\times$, we may form the quotient group

$$\Gamma = K^\times / U.$$

This is called the *value group*.

Each $x \in K^\times$ determines a class

$$\overline{x} \in \Gamma.$$

Meaning. Two elements $x, y \in K^\times$ define the same class if and only if

$$xy^{-1} \in U,$$

that is, if they differ by a unit. Thus Γ forgets the unit part and remembers only the essential "order" or "size" of an element.

9.12 12. Ordering the value group

Define an order on Γ by

$$\overline{x} \geq \overline{y} \iff xy^{-1} \in A.$$

Because A is a valuation ring, for any $x, y \in K^\times$, either

$$xy^{-1} \in A \quad \text{or} \quad yx^{-1} \in A.$$

So this order is total.

Moreover, it is compatible with the group structure. Thus Γ becomes a totally ordered abelian group.

Example: $k((t))$. If

$$x = t^m u, \qquad y = t^n v,$$

with $u, v \in k[[t]]^\times$, then

$$xy^{-1} = t^{m-n}(uv^{-1}).$$

This lies in $k[[t]]$ if and only if $m - n \geq 0$. Hence

$$\overline{x} \geq \overline{y} \iff m \geq n.$$

So in this case

$$\Gamma \cong \mathbb{Z}$$

with its usual order.

9.13 13. The valuation map

Define

$$v : K^\times \to \Gamma, \qquad v(x) = \overline{x}.$$

This is the valuation map in multiplicative notation. Often one extends it by setting

$$v(0) = \infty.$$

If one writes the group law on Γ additively, then:

$$v(xy) = v(x) + v(y).$$

The key inequality is:

$$v(x+y) \geq \min(v(x), v(y)) \qquad \text{whenever } x, y \in K^\times,\ x+y \neq 0.$$

9.14 14. Proof idea of the valuation inequality

Assume

$$v(x) \geq v(y).$$

By definition this means

$$xy^{-1} \in A.$$

Then

$$\frac{x+y}{y} = xy^{-1} + 1 \in A,$$

since both xy^{-1} and 1 lie in A. Therefore

$$(x+y)y^{-1} \in A,$$

which exactly means

$$v(x+y) \geq v(y) = \min(v(x), v(y)).$$

The case $v(y) \geq v(x)$ is symmetric.

9.15 15. Geometric and arithmetic intuition

Valuations measure vanishing, poles, or divisibility.

Examples.

- In $k(x)$, the $x = 0$ valuation measures order of zero or pole at 0.
- In $k((t))$, the valuation measures the first exponent of t.
- In number theory, the p-adic valuation measures divisibility by p.

Thus:

- the valuation ring consists of elements of nonnegative order,
- its maximal ideal consists of elements of strictly positive order,
- the units are exactly the elements of order 0.

9.16 16. Comparison table

Local ring. A ring with exactly one maximal ideal.

Example:

$$k[x, y]_{(x,y)}.$$

Valuation ring. A local subring $A \subseteq K$ such that for every $x \in K^\times$,

$$x \in A \quad \text{or} \quad x^{-1} \in A.$$

Example:

$$k[[t]] \subset k((t)).$$

DVR. A valuation ring whose value group is isomorphic to $\mathbb{Z}$.

Examples:

$$k[[t]], \qquad \mathbb{Z}_{(p)}, \qquad k[x]_{(x)}.$$

9.17 17. Most important examples to remember

Example A: $k[[t]] \subset k((t))$.

- valuation ring,
- local,
- maximal ideal (t),
- value group $\mathbb{Z}$.

Example B: $\mathbb{Z}_{(p)} \subset \mathbb{Q}$.

- valuation ring,
- local,
- maximal ideal $p\mathbb{Z}_{(p)}$,
- value group $\mathbb{Z}$.

Example C: $k[x, y]_{(x,y)} \subset k(x, y)$.

- local,
- not a valuation ring.

This last example is the standard counterexample showing that local rings are more general than valuation rings.

9.18 18. Final summary

- A *local ring* is a ring with one maximal ideal.
- A *valuation ring* is a local subring of a field such that every field element or its inverse lies in the ring.
- Every valuation ring is local, but not every local ring is a valuation ring.
- Valuation rings are exactly the maximal local subrings of a field under domination.
- Their unit group U gives the value group

$$\Gamma = K^\times/U,$$

which is a totally ordered abelian group.

- The valuation map
$$v : K^\times \to \Gamma$$
satisfies
$$v(xy) = v(x) + v(y), \qquad v(x + y) \geq \min(v(x), v(y)).$$